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Emotional Impact on Health Care Providers Involved in Medical Assistance in Dying (MAiD): A systematic review and qualitative meta-synthesis.

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Title: Emotional Impact on Health Care Providers involved in Medical Assistance in Dying (MAiD): A systematic review and qualitative meta-synthesis.

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Abstract:

Objective: To determine the emotional impact on health care providers involved in Medical Assistance in Dying.

Introduction: Medical Assistance in Dying (MAiD) traverses challenging and emotionally overwhelming territories: Health Care Providers (HCPs) across jurisdictions bridge the divide between normative yet opposing values of sanctity of life and dignity in death and dying resulting in a myriad of affective responses. These range from a rewarding experience on one end to an overwhelming sense of apprehension and unpreparedness on the other.

Methods: A systematic review research methodology was adopted to review qualitative research studies from 4 databases (OVID Medline, EMBASE, CINAHL, and Scopus) and grey literature. Key author, citation, and reference searches were also undertaken. Papers were included if they presented qualitative data regarding the emotional impact on HCPs involved in MAiD. Studies were restricted to English language. Analysis was conducted using thematic meta-synthesis. Once thematic synthesis was completed, the cumulative evidence was assessed using the Confidence in the Evidence from reviews of qualitative research (CERQual) approach.

Results: The search identified 4523 papers. After applying inclusion/exclusion criteria and The Joanna Briggs Institute Critical Appraisal Tool for qualitative research, 35 papers were included in the meta-synthesis. Three distinct emotional themes were identified— (1) Strong, internalized basic emotions including moral distress, (2) Role-based emotions based on individual personal/moral/professional values, and (3) Reflective emotions that point towards MAiD being a 'sense-making process'.

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Conclusion: This systematic review and meta-synthesis maps the current emotional discourse amongst HCPs involved in MAiD and discusses the role of legal frameworks and the position of the health care provider in shaping this discourse. It attempts to answer the question of 'what it means at an emotional level' for a healthcare provider to engage in MAiD.

Key words: Ethics-Medical, Health Personnel, Euthanasia, Right to Die.

Strengths and Limitations of this study:

Strengths:

- An eligibility criteria and subsequent search strategy that focusses on emotional impact of MAiD on HCPs with qualitative research methodology.
- Use of Joanne Brigg's critical appraisal tool for assessment of risk of bias and use of the CERQual approach for assessing the methodological limitations, relevance, coherence and adequacy of the evidence after completion of meta-synthesis.

Limitations:

- Qualitative signals of absence of sub-group analysis, eligibility criteria limited to published English-language literature and fast-moving pace of research on emotional impact of MAiD on HCPs likely contributes to significant publication bias.
- Generalizability of evidence limited by presence of selection bias in included studies.

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Introduction:

Medical Assistance in Dying (MAiD) poses ethically complex challenges that can be a major source of distress to Health Care Providers (HCPs) who choose to participate—especially since MAiD involves navigating conflicting personal and professional values. These values are contextual, dynamic and often not in alignment with each other; for example, duty to care and reducing suffering in case of terminal illness through MAiD is in sharp contrast with the value of preserving sanctity of human life in the dying process (1, 2). Except for Switzerland, all other countries require HCPs to be at the forefront in discussing and executing eligible requests for assisted death within their defined jurisdictions (3).

Assisted death in selected jurisdictions-overview and current status

The number of jurisdictions across the world with medically assisted death legislation continues to grow. Switzerland, Netherlands, Belgium, Luxemburg, Canada, besides jurisdictions in the USA (Oregon, Vermont, California, Washington State, Colorado, the district of Columbia, Hawaii, Maine and New Jersey) along side the State of Victoria, Tasmania and South Australia in Australia and Columbia in South America, and most recently Spain and New Zealand, have legalized medically assisted death in some form (3, 4). MAiD in Canada, the State of Victoria in Australia and the Benelux countries includes both assisted suicide and euthanasia. Jurisdictions in the USA and Switzerland allow only assisted suicide.

Broadly speaking, the 'Benelux' countries (Belgium, Netherlands and Luxemburg) have less restrictive rules in place for MAiD than the American jurisdictions that permit this practice. For example, Benelux countries allow advanced directives and terminality of illness is not a requirement to be eligible for MAiD in Belgium and Netherlands. Jurisdictions in the USA, on the other end, have strict eligibility criteria that the illness must be terminal and there must be some timeline to foreseeability of natural death—commonly 6 months in most jurisdictions.

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Right to bodily autonomy, voluntariness of request and terminality/irremediableness of the medical condition are the mainstay of the eligibility criteria for MAiD, with each criterion receiving variable emphasis, depending on the legislative jurisdiction. For example, "reasonable foreseeability of natural death" criterion was removed from Canada's MAiD eligibility criteria following recent changes in the legislation (5-8).

HCPs and MAiD—current knowledge and knowledge gaps.

Amongst the HCPs, the physician's role in providing MAiD is perhaps the most ambiguous. Historically, medicine as a profession is rooted in the ethical principle of 'first, do no harm' while providing care. While this is true, medical futility and the sense of powerlessness and loss of control at end-of-life are a reality in modern medical practice, which is often reflected as physician ambivalence to participate in MAiD (9-11).

While this sense of ambiguity and a morally contradictory stance distances physicians from the practice of MAiD, nurses also share the complex attitudes and polarized feelings towards MAiD (12). This complexity is often due to the dual role that nurses play in most health care systems around the world: on one end, they act as a strong advocate for patient's wishes, whereas on the other end, they only have a supportive role in medical decision-making process. A recent synthesis of qualitative studies describing registered nurses' experiences with MAiD from Belgium, the Netherlands, and Canada showed that while the nurses played a central role in providing important 'wrap-around' care for patients and family, their participation in MAiD required significant moral work (13).

A recent scoping review exploring the challenges faced by HCPs while handling MAiD requests found lack of clear guidelines/protocols, role ambiguity, difficulties in evaluating capacity/consent, conscientious objection, lack of inter-professional collaboration and difficulties in assessing nature and severity of suffering as major barriers in developing comprehensive care models for implementation of MAiD (14).

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Furthermore, the scoping review also pointed out that HCPs need substantial degree of time and emotional commitment to participate in a MAiD request. A scoping review and thematic meta-synthesis of qualitative studies exploring HCPs' attitudes towards assisted death practices in Belgium, Netherlands, Israel, Australia, Germany and the USA showed that their attitudes were shaped by a deep sense of moral responsibility and contextual care-relationships (15).

This empirical evidence provides valuable insights on experiences and attitudes of HCPs towards MAiD; however, the nature and extent of emotional impact remains unexplored. Preliminary evidence suggests that the shift away from the medicine's fundamental values of cure/control of a disease process to hastening death with the aim of alleviating intolerable suffering can have significant emotional effects on participating HCPs. These can range from feeling overwhelmed with a sense of powerlessness on one end, to a rewarding and a positive experience on the other (16, 17).

Objectives: To determine the emotional impact on HCPs involved in MAiD.

Methods:

Search strategy, screening and eligibility criteria:

The inclusion and exclusion criteria were developed in line with SPIDER (Sample, Phenomenon of Interest, Design, Evaluation, Research type) (18). We included all qualitative research studies that evaluated the emotional impact of MAiD on HCPs. In order to ensure qualitative richness of themes, we excluded all surveys, personal anecdotes, experiences without in-depth qualitative analysis, opinions, attitudes or comments published on this topic.

Relevant definitions: For the sake of this review, we define a Health Care Provider as a person "lawfully entitled under the law of a province to provide health services in the place in which the services are provided by that person"(19). This definition includes pharmacists, nurses, nurse practitioners, social

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workers, spiritual health practitioners, psychotherapists and clinical psychologists who are legally authorized to practice within their respective scope of practice. For the sake of this study, we included 'Assisted suicide assistant' and provider in 'Right to die' societies in Switzerland as unique MAiD care-providers who contact the eligible participant and liaise with the physician and pharmacist in the conduct of MAiD.

For the sake of this review, the term 'MAiD' refers to (20):

a. The administering by a physician or nurse practitioner of a substance to a person, at their request, that causes their death (euthanasia); and/ or

b. The prescribing or providing by a physician or nurse practitioner of a substance to a person at their request, so that they may self-administer the substance and in doing so cause, their own death (assisted suicide).

Eligibility criteria:

1) Includes worldwide published literature on the research question in English language, inclusive of all age groups; articles published up to April 30, 2021 were included.

2) Describes or mentions 'HCPs' and 'MAiD' as defined above

3) Describes or mentions the emotional impact on HCPs in terms of emotions /affective responses experienced or expressed while accessing, discussing, participating or caring for the patient who has made a valid MAiD request.

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4) Includes all qualitative studies evaluating the emotional impact through qualitative research methodologies like grounded theory, semi-structure interviews, lived experiences, phenomenology, thematic analysis, narrative inquiry or others.

5) Excludes case studies, anecdotes or studies without a description or mention of a rigorous qualitative research methodology.

Search strategy:

An iteratively developed search strategy was developed and piloted with the help of 3 librarians with expertise in systematic review search strategies. Considering the inter-disciplinary nature of the objective, the search strategy was conducted on OVID Medline (to cover all the North American literature), CINAHL (to cover all the nursing literature), EMBASE (to cover all the European literature) and SCOPUS (to cover all miscellaneous literature, presentation abstracts, etc.). The search terms included three main domains—MAiD, HCPs and qualitative research methodology and their synonyms.

A combination of subject heading and key word searches were trialed to ensure a balance between sensitivity and specificity. Full search strategy on the OVID medline database as reference is available in supplementary appendix 1.

In addition to database searches, the study team conducted a grey literature search (21) which was informed by search methods outlined by Godin et al (22). The search terms included MAiD, qualitative research, HCPs, and their synonyms. Grey literature was retrieved between December 10, 2018 and March 1, 2019 and updated on August 10, 2020 and August 10, 2021 from:

(1) Databases including Google scholar, the Canadian electronic library and the Canadian Institute for Health Information and

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(2) OpenGrey, BASE (Bielefeld Academic Search Engine) and the OAIster catalogue of open access resources that includes digital thesis sources like the WorldCat.

The grey literature search strategy and results are included in supplementary appendix 1. For the purpose of feasibility and relevance, only reports from the year 2000 and beyond were retrieved. In addition, backward citation tracking was conducted by hand searching the reference lists of all included papers.

Study selection process:

All identified records were imported into the reference management software, Zotero and duplicates removed by the lead researcher (SD). 20% of the title and abstracts of peer reviewed records were independently screened by two reviewers (AS and AB) based on the eligibility criteria. Given that a substantial portion of grey literature did not include abstracts, the grey literature screening process was initiated at the full-text phase. Records were excluded if they did not follow a qualitative research methodology and were not peer reviewed. SD consulted the keywords of yielded academic records if the title and abstract lacked clarity in relation to core concepts and reviewers AB and AS independently assessed any records for which there was a discrepancy and/or uncertainty regarding their inclusion. The researchers met at the beginning, middle and end of the screening process to ensure consistency. The same inclusion and exclusion criteria, successive team meetings and approach to discrepancies was applied to second level screening of the full-texts of the academic literature, as well as the full-texts of the grey literature.

Patient and Public involvement: No patients involved.

Assessment of risk of bias:

We used the Joanna Briggs Institute Critical appraisal tool for use in systematic reviews: checklist for qualitative research to critically appraise the included full-text qualitative research studies over 10

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constructs. These constructs range from congruency to philosophical construct to theoretical and cultural location of the researcher (23). The results of the assessment of risk of bias were independently reviewed by AB and AS and are presented in detail in supplementary appendix 2.

The search results and reasons for exclusion at each stage of screening were recorded and are represented in the adapted Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) flow diagram in Figure 1.

Data Analysis:

Data Extraction and Data analysis:

We adopted a Thematic Synthesis Approach to analyze and synthesize data. Thematic synthesis is an adaptation of thematic analysis and provides a set of established methods and techniques that help synthesize qualitative research outcomes, especially when there is heterogeneity in the outcome variables (24). A Thematic Synthesis Approach is especially useful in our case since it enables us to examine the meaning, significance and social constructions around the emotional experience of a healthcare provider involved in MAiD. The coded data were sent to AB and AS to cross check for any discrepancy. Subsequent thematic synthesis was done by SD, AB and AS in the following 2 stages:

Stage 1: Identifying the similarities between the codes.

All relevant qualitative data from the selected primary studies were extracted manually from the results, discussion and conclusion section of individual studies and are represented in Table 2 of supplementary appendix 2. The codes (done line-by-line to search for concepts) were grouped into descriptive themes inductively so that patterns could be identified. Each theme was entered as boxes and codes from each study illustrated in those boxes, so that constant comparison analysis process could be done (see Table 3 in supplementary appendix 3).

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Stage 2: Development of analytic themes.

In this last stage, meaning of the patterns was analyzed against the research question so that a narrative component could be developed.

Once thematic synthesis was completed, each researcher independently evaluated the cumulative evidence from individual studies for methodological limitations, relevance, coherence and adequacy using the Confidence in the Evidence from reviews of qualitative research (CERQual) approach (see table 1) (25).

All researchers met during regular research-review meetings to resolve any discrepancies and achieve consensus over the assessment.

This systematic review was a part of an academic capstone project and was not registered with any international database. The review protocol is available from the research team on request.

In addition to employing the PRISMA Checklist for systematic reviews, in order to improve the reporting of our qualitative meta-synthesis, we use the Enhancing transparency in reporting the synthesis of qualitative research (ENTREQ) checklist which is accessible from supplementary appendix 4.

Results:

Characteristics of included studies:

35 qualitative research studies were included in the review. The included literature was based in 5 countries: The United States of America (7), The Netherlands (9), Canada (14), Belgium (1), Switzerland (3), and one study was an international study with participants from the United States of America and Netherlands. The data included 393 physicians, 169 nurses, 53 social workers in hospice care, 11 allied health care professionals (7 personal support workers, 1 pharmacist, 1 genetic technologist and 2 psychologists) and 8 directors of socio-medical institutions and 3 socio-cultural animators. A detailed description of the included studies is included in Table 2 of supplementary appendix 3.

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Thematic synthesis*:*

Three overarching emotional dimensions were derived from the thematic synthesis:

Dimension 1: Strong, internalized and polarized emotions (26-36): These included three subordinate categories/genres:

- Positive emotions of 'reward', 'relief', 'active openness', 'overwhelming but uplifting' feelings while participating in MAiD,
- Negative emotions of 'powerlessness', 'guilt', 'emotional exhaustion', 'vicarious suffering' and fear of a slippery slope and losing control, and
- Individual conscience-based emotions of 'moral shudder' on one end and feelings of 'mercy' on the other end while participating in MAiD.

This emotional dimension was strongly embedded in the cultural and political milieu and the interpersonal communication strategies used by the HCP.

Dimension 2: Reflective, discourse-based emotions (26,30,36,37-45): These included emotions of 'growing with the patient's experience', MAiD as a 'sense-making process', 'de-tabooing the philosophical meaning of death through MAiD' and various degrees of 'dynamic conflict' secondary to a reflective sense of insecurity. These emotions were descriptively laid on a platform of 'interpretative therapeutic engagement', where they seemed to aid in the larger philosophical and societal discourse around MAiD (46).

Dimension 3: Emotions that resonate with professional values and/or legislative frameworks (28,30,34, 47-61): These included emotions embedded in and modulated by the HCPs professional and legal milieu. They resonated with professional values like 'competency and perfection', 'intimate care', '*colloque*

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singuliar' (singular language of trust and conscience in context of therapeutic relationship) and various degrees of commitment ranging from 'contractual' to 'sacrificial'.

In order to explore how HCPs represent themselves, or their emotions, to themselves and to larger health care environment, we adopted a narrative inquiry approach. This allowed us to extract how and why did the HCPs participating in MAiD experience such complex emotions. While the thematic synthesis focused on broad aspect of the 3 dimensions of emotional impact described above, the narrative inquiry approach focussed on contextual factors leading to the emotional impact. Based on the narrative inquiry approach, a narrative summary was formulated and is described below:

1. In jurisdictions that legislate MAiD with the central aim to alleviate intolerable suffering in context of terminally ill medical conditions (example the USA), the HCPs experience strong polarized emotions that are modulated by their individual cultural/religious background. The extent of emotional impact ranges from positive emotions of reward/relief on one end, to negative (burden, emotional exhaustion) and conscientious based moral distress on the other.
2. In jurisdictions that legislate MAiD with an emphasis on alleviating intolerable suffering without terminal illness being a necessary requirement (for example Benelux countries, Switzerland, Canada), the HCPs experience the emotional impact of MAiD as a 'sense-making' process—this allows them to reflect on the emotional dissonance between basic emotions and emotions that conform to legislative rules.
3. Values associated with the health care provider's profession (example, physician, nurse or social worker), their degree of engagement in the MAiD process, depending on their hierarchical position in the health care system and legislations of respective jurisdictions are strong influential factors that shape the emotional impact of MAiD.

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Table 1: Grading of Recommendations, assessment, development and evaluation (GRADE) Confidence in the evidence of reviews of Qualitative Research (CERQual) evidence profile

Summary finding	Studies contributing substantially to the summary theme (studies numbered as per Table 2 in supplementary appendix 3)	Methodological Limitations	Coherence	Adequacy	Relevance	CERQual GRADE evidence	Explanation of CERQual assessment
HCPs experienced strong, internalized, often polarized and deeply	1,2,3,5,13,18, 19,21, 24,26,28.	Minor methodological limitations concerning location of the researcher	Moderate concerns regarding coherence	Minor concerns regarding adequacy	No or very minor concerns regarding relevance	High	Variability in experiences of participants posed a challenge with respect to

On applying the GRADE CERQual approach to ascertain the degree of confidence in these findings, we graded the evidence in terms of adequacy, relevance, coherence as well as methodological limitations (25).

While evidence supporting all three dimensions of emotional impact had methodological limitations, evidence for emotions shaped by professional values and corresponding legislative boundaries had significant selection bias leading to lack of generalizability. Variability in experiences of participants posed a challenge with respect to coherence, especially in studies that reflected emotional impact driven by HCPs' cultural and religious background; however, this variance enhanced the richness of the results. The evidence from studies describing the emotional impact on HCPs due to the socio-political environment in which MAiD was practiced showed good coherence, adequacy and relevance; however, it was understandably influenced by the position of the researcher. A detailed account of the summary of findings is described in Table 1.

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personal basic emotions that were modulated by the HCP's cultural and/or religious background. **Level embedded**: cultural/religious		theoretically/ culturally, And influence of the researcher on the research and vice versa					coherence, however, this also added to the richness of results. Hence, we have graded the confidence in quality of findings as high.
Influenced by the socio-political environment as well as the social discourse on suffering and death, HCPs shared emotions of personal growth/sense-making and relational experiences of deeper compassion and sympathy. HCPs also experienced emotional dissonance over personal emotions and emotions expressed to conform to legislative rules. **Level embedded**: Socio-political	2,5,6,8,11,14, 23,25, 28,30, 32, 34.	Moderate/min or methodological limitations concerning location of the researcher theoretically/ culturally, and influence of the researcher on the research and vice versa	No or very minor concerns regarding coherence	No or very minor concerns regarding adequacy	No or very minor concerns regarding relevance.	High	Paper 6 did not approach the ethics committee and hence does not have ethics committee approval. Apart from this study, all studies in this group contributed to the summary findings in terms of coherence, adequacy and relevance. Hence, we have graded the confidence in the quality of the findings to be high.

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HCPs expressed emotions aligned with their individual professional values and beliefs systems and, most of the times, attempted to align their values associated with the MAiD ideology; at other times, legislation of respective jurisdictions helped shape emotional experiences. **Level embedded**: Professional/ legal	3,4,5,7,9,10,12, 15,16, 17,20,22,24,27, 29,31, 33,35.	Moderate Methodological limitations concerning location of the researcher theoretically/culturally, And influence of the researcher on the research and vice versa. Also, selection of participants Paper 16, one single hospital.	minor concerns regarding coherence	No or very minor concerns regarding adequacy	No or very minor concerns regarding relevance	Moderate	Most of the studies in this group had methodological problems of selection bias and lack of generalizability. For example, paper 16 selected participants from a single hospital-based setting. The findings are limited in terms of generalizability to similar groups in different settings. Hence, we have graded down our confidence in the quality of findings to moderate.

Discussion:

This systematic review and thematic meta-synthesis attempts to answer the question of 'what it means at an emotional level' for HCPs involved in providing MAiD.

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Discourse on emotional Impact of MAiD in Benelux vs. Non-Benelux countries—key features:

The substantive and procedural requirements for MAiD across global jurisdictions rests on 3 main pillars: patients' right for self-determination and respect for bodily autonomy expressed through voluntariness of request and a valid, informed consent process, foreseeableness of natural death due to terminal medical illness and subjective nature of individual suffering (62,63). The key difference between the legislations for MAiD in Benelux countries and countries like the USA is the differential emphasis on eminent or foreseeableness of death. The MAiD legislations in Belgium, Netherlands, Switzerland, and Canada have a more permissive legal framework that allows people to access MAiD as a service to end their intolerable suffering that has no prospect of improvement but is not necessarily terminal.

On one end, attitudes of physicians towards MAiD has shown reflective trends to legislative standards; countries like Belgium and Netherlands find much stronger physician support than their USA counter parts (64). On the other end, public support towards MAiD has been reflective of the prevailing cultural and religious practices; central and eastern European countries have shown a decline in support with corresponding increase in religiosity as opposed to western European countries (65,66). The attempt by HCPs to align themselves with their own individual values, legislative standards and public perceptions while engaging in MAiD can lead to intense emotional responses, both, within their internal, personal and their external professional spaces.

An important take home message from this research is how legislations have a shaping effect on emotional responses. The HCPs who practice in the Benelux countries and Switzerland seem to experience reflective emotions over strong polarizing emotions expressed by HCPs who practice in non-Benelux countries like the USA. Canada seems to have a unique, transitional position—with the emphasis of the legislation going the Benelux countries' way, the HCPs emotional experiences show a mixture of emotions driven by their professional values as well as the ongoing societal discourse on MAiD. This observation

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conforms to Michel Foucault's position on how law acts as an element in the expansion of power(s) (67); legislatures along with other platforms of knowledge expression modulate every fiber of human society. Our narrative synthesis points out that the Law that limits application of MAiD to terminally illnesses provide for a more broad range of emotional expression. Thus, legislation on MAiD across the globe provides the HCP with a locus of administrative control which then decides how the emotional discourse around MAiD is shaped; the question is—*how* do we want the *emotional discourse* around MAiD to be shaped?

Emotional discourse amongst HCPs involved in MAiD: position of the HCP and ethics of Care

The right to choose when and how to die has always been a contentious issue across various societies (68-70). Public perceptions on MAiD are shaped through societal emphasis on individual as well as contextual factors associated with assisted death—these are often linked with sense of identity, awareness of personal pain and suffering, religious beliefs regarding sanctity of human life and personal meaning of death, and loss of autonomy and dependence associated with illness-related intolerable suffering. With advancing medical technologies, the potential to prolong life has increased significantly (71,72), and the HCPs assumes a central position to shape the discourse around assisted death.

In countries where MAiD is legalized but is restricted to terminal illnesses with imminent chance of death, the position of a HCP continues to be one that of a provider of 'Care'. Here, the moral dimension of 'Care' continue to be defined as 'everything we do to maintain, continue or repair our world so that we can live in it as well as possible' (73). The Value of Care in context of health care have always been traditionally associated with attentiveness, responsibility, nurturance, compassion and meeting others' needs (74). While emotional responses to legal requests of hastening death is affected by policies, professional identity, commitment to patient autonomy, personal values and beliefs, the patient-clinician relationship and will vary on a case-by-case basis (75); this systematic review raises an important question—How does

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legalizing MAiD with emphasis on alleviating intolerable suffering without the context of a terminal illness change the moral dimensions of Care?

Conclusion:

The myriad of emotions experienced by HCPs participating in MAiD are influenced by their individual socio-cultural values, professional role and position in the health care system and the legal framework under which they practice MAiD. This emotional discourse is rich and diverse; HCPs experience dimensions of strong positive/negative emotions, reflective, sense-making emotions and/or professional value driven emotions. HCPs practicing MAiD under jurisdictions that require terminal illness as an essential criterion experience more polarized, positive/negative emotion. HCPs practicing in jurisdictions that do not require this as an essential criterion but are legislated with greater emphasis on allaying intolerable suffering experience more reflective emotions driven by the larger societal discourse around MAiD.

Limitations of the review:

This review is limited by its focus of emotional impact on HCPs only and the obvious selection bias in the included studies—those who could and volunteered to express their emotions are represented in the review. The review is also limited with absence of sub-group analysis with respect to HCPs' age, years of experience and the influence of gender on the emotional discourse on MAiD. In addition, although our search strategy does include specialized bibliographic databases and an extensive grey literature search, inclusion of only English language studies likely points towards high risk of publication bias.

There are several gaps in our understanding of the emotional impact on HCPs involved in MAiD that would benefit from further research. Unbearable or intolerable suffering is the driving force for patients

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requesting MAiD, and an empathetic understanding of suffering regulates the unique emotional experience of a HCP providing MAiD. An in-depth exploration of this nebulous concept of intolerable suffering in context of MAiD may help HCPs navigate their emotional experience while providing MAiD.

Ethics statement: This is a systematic review and meta-synthesis of already published and accessible research data and does not require ethics committee or Institutional board approval.

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Contributorship Statement: The authors confirm contribution to this systematic review and meta-synthesis as follows:

- study conception and design: Dr. Saumil Dholakia, Dr. Alireza Bagheri, Dr. Alexander Simpson.
- development of eligibility criteria: Dr. Saumil Dholakia, Dr. Alireza Bagheri and Dr. Alexander Simpson.
- search strategy developed by Erica Lenton, Heather Cunningham (Library services, Gerstein's library, University of Toronto) and Fiona Inglis (Library services, Center for Addiction and Mental Health, University of Toronto) in close consultation with Dr. Saumil Dholakia and reviewed by Dr. Alireza Bagheri and Dr. Alexander Simpson.
- Study selection and data extraction process by Dr. Saumil Dholakia and independently reviewed by Dr. Alireza Bagheri and Dr. Alexander Simpson.
- Dr. Saumil Dholakia performed the assessment of risk of bias, which was independently reviewed by Dr. Alireza Bagheri and Dr. Alexander Simpson.

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- All three authors were involved equally in performing the qualitative meta-synthesis and CERQual assessment.
- Draft manuscript preparation: Dr. Saumil Dholakia with multiple reviews, feedback and edits in form as well as content by Dr. Alireza Bagheri and Dr. Alexander Simpson.
- All authors reviewed the results and approved the final version of the manuscript.

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Figure 1: PRISMA flow diagram: The PRIMSA diagram details our search and selection process applied during the review.

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Figure 1: PRISMA flow diagram:

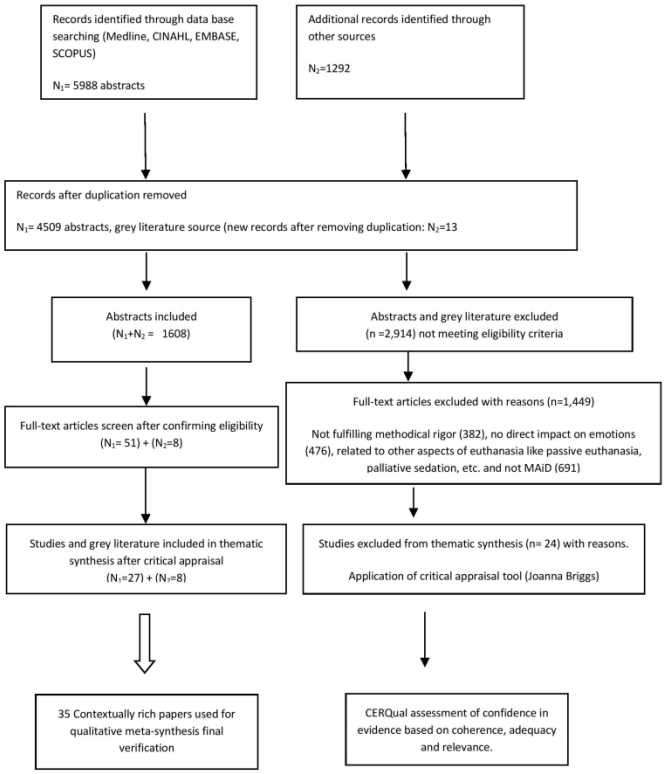


Figure 1: PRISMA flow diagram: The PRIMSA diagram details our search and selection process applied during the review.

215x279mm (300 x 300 DPI)

Supplementary appendix 1:

Database: Ovid MEDLINE(R) ALL <1946 to October 04, 2021>
Search Strategy:
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1 euthanasia, active/ or euthanasia, active, voluntary/ or hospice care/ or suicide, assisted/ (14808)
2 terminally ill/ (6684)
3 Right to die/ (4950)
4 Terminal care/ (29907)
5 advance care planning/ or advance directives/ (9125)
6 ((dying or death or euthan* or suicide or terminal* ill*) adj5 (assist* or hasten*)).tw,kf. (5952)
7 Palliative care/ (58012)
8 exp Practice Patterns, Physicians'/es [Ethics] (812)
9 physician's role/ (30584)
10 Health Personnel/ (52294)
11 ((health care provider or clinician* or doctor* or physician* or nurse or social work* or oncologist* or palliative physician or nursing or psychiatrist* or psychologist* or psychotherapist*) adj3 (experience* or emotion* or feeling*)).tw,kf. (23976)
12 (Interview: or experience:).mp. or qualitative.tw. (1655368)
13 health personnel/ or allied health personnel/ or anesthetists/ or caregivers/ or case managers/ or "coroners and medical examiners"/ or emergency medical dispatcher/ or epidemiologists/ or faculty, medical/ or faculty, nursing/ or health educators/ or health facility administrators/ or medical chaperones/ or medical laboratory personnel/ or medical staff/ or exp nurses/ or nursing staff/ or occupational therapists/ or personnel, hospital/ or pharmacists/ or physical therapists/ or physician executives/ or exp physicians/ (431546)
14 (ethnograph* or grounded theory or qualitative research or thematic analysis or semi-structured interview* or narrative inquiry or focus* group or content analysis or discourse or lived life experience*).tw,kf. (156494)
15 aid in dying.mp. (243)
16 death with dignity.mp. (607)
17 Bill C-14.mp. (24)
18 Bill C-7.mp. (2)
19 MAID.mp. (458)
20 physician assisted death.mp. (309)
21 physician assisted dying.mp. (142)
22 (assisted suicide or physician assisted suicide).tw,kf. (3163)
23 Qualitative Research/ (67825)
24 1 or 2 or 3 or 4 or 5 or 6 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 (55486)
25 7 or 8 or 9 or 10 or 11 or 13 (527655)
26 12 or 14 or 23 (1692068)
27 24 and 25 and 26 (5490)
28 limit 27 to dt=19460101-20210430 [January 1st, 1946 to April 30th, 2021] (5441)
29 limit 28 to english language (5073)
30 limit 29 to abstracts (4876)

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Grey Literature databases (December 10th 2018 to March 1st, 2019, updated August 2020 and 2021)

Total # Records after de-duplication: 13. Records selected after applying critical appraisal tool 8.

Database	Search strategy	#records screened	# new records and records after de-duplication and applying the critical appraisal tool
Google scholar	With the exact phrase: "Medical assistance in dying" ; "physician assisted suicide"; With all the words: "emotional impact on health care providers involved in medical assistance in dying"	400	5
Des Lebris/Canadian Electronic Library	Medical assistance in dying	5	0
Canadian Institute of Health Information (CIHI)	Medical assistance in dying	7	0
OAIster database (includes WordCAT)	Medical Assistance in dying, Physician assisted suicide as key word	206	2
OpenGrey	Medical assistance in dying, Physician Assisted suicide as key word	4	0
BASE (Bielefeld Academic Search Engine)	Subject Heading search: "Medical Assistance in dying"	670	1

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Selected records:

Google scholar included Results:

1. Beuthin R. Cultivating Compassion: The Practice Experience of a Medical Assistance in Dying Coordinator in Canada. *Qualitative Health Research*. 2018;28(11):1679-1691. doi:10.1177/1049732318788850
2. Bouthillier M-E, Opatrny L. A qualitative study of physicians' conscientious objections to medical aid in dying. *Palliative Medicine*. 2019;33(9):1212-1220. doi:10.1177/0269216319861921
3. Bruce A, Beuthin R. Medically Assisted Dying in Canada: "Beautiful Death" Is Transforming Nurses' Experiences of Suffering. *Canadian Journal of Nursing Research*. June 2019. doi:10.1177/0844562119856234
4. Alison A. Townsley. Putting a Bow on Death and Dying-Health Care Professionals' Experiences with Medical Assistance in Dying (MAiD): A Foucauldian Discourse Analysis with Agambian Insights. [Practice Based Research Paper on the Internet]. York University School of Social Work; 2018. [Cited February 28,2019] Available from: https://yorkspace.library.yorku.ca/xmlui/bitstream/handle/10315/35613/Alison_Townsley_PRP_2018.pdf?sequence=1
5. Gamondi C, Borasio GD, Oliver P, Preston N, Payne S. Responses to assisted suicide requests: an interview study with Swiss palliative care physicians. BMJ Support Palliat Care. 2019 Mar;9(1):e7. doi: 10.1136/bmjspcare-2016-001291. Epub 2017 Aug 11. Available from https://pubmed.ncbi.nlm.nih.gov/28801317/

OAIster included Results:

1. Implementing Medical Aid-in-Dying in Vermont: a Qualitative Study. *J GEN INTERN MED* **34,** 636–641 (2019). https://doi.org/10.1007/s11606-018-4811-1
2. Allyson Oliphant. "If Not Me, Then Who?" The Narratives of Medical Aid in Dying (MAiD) Providers and Supporters Around Their Professional Identity and Role in MAiD [Thesis on the Internet]. MacMaster University; 2017. [Cited on February 10, 2019]. Available from http://hdl.handle.net/11375/22146

BASE included results:

1. Sheridan, Laura, "When Patients Ask to Die: The Role of Nurses in Medical Assistance in Dying" (2017). Electronic Thesis and Dissertation Repository. 5041. https://ir.lib.uwo.ca/etd/5041

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Study (location, number and category of participants)	JBI Questionnaire	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	Appraisal	R for exclusion
1.Voorhees et al., US and Netherlands, physicians 23		Y	Y	Y	Y	Y	Y	N	Y	Y	Y	Include	
2.Van Marwjik et al. Netherlands 22 Primary care ph		Y	Y	Y	Y	Y	N	N	Y	Unc	Y	Include	
3. Denier Yyonne et al. 2010. Belgium Nurses-n=18		Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
4. Elizabeth Norton et al. 2012 USA-social worker-9		Y	Y	Y	Y	Y	N	N	Y	Unc	Y	include	
5. JJ Georges et al. 2008. Netherlands 30 GPs		Y	Y	Y	Y	Y	N	N	Y	Unc	Y	Include	
6. Snijdewind et al., 2014 (Netherlands, 28 physicians)		Y	Y	Y	Y	Y	N	N	Y	N	Y	include	
7. Katja ten Cate et al., 2017-33 physicians netherlands		Y	Y	Y	Y	Y	N	N	Y	Y	Y	Include	
8. Donald G Van tol et al., 2012. Netherlands 15 physicians		Y	Y	Y	Y	Y	N	N	Y	N	Y	include	
9.Veronica Lorraine Fausto Melchor, 2018. USA Hospice social worker 8		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	include	
10. Pamela Miller et al., 2008 Oregon SW-8		Y	Y	Y	Y	Y	N	N	Y	Unc	Y	Include	
11. Deborah Volker et al., 2001. USA Onco Nurse-40		Y	Y	Y	Y	Y	N	N	Y	Unc	Y	Include	

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12.michael Young et al., 2008 Canada nurses-22	Y	N	Y	Y	Y	N	N	Y	Y	Y	**exclude**	Study done at a time assisted death not legal, so does not meet inclusion criteria.
13. Rosanne Beuthin et al., 2018 Canada nurses-17	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
14.eva Bolt et al., 2017 Netherlands paediatrician-8	Y	Y	Y	Y	Y	N	N	Y	NR	Y	Include	
15.Dolores Angela Castelli Dransart et al., 2017 Switzerland-20 nurse, 1 physician, 8 directors, 3 socio-cultural animators.	Y	Y	Y	Y	Y	N	N	Y	Y	Y	Include	
16. Marianne Dees et al., 2012 Netherlands-phy-28	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
17. Theresa Harvath et al., 2006. USA hospice social workers-20	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
18. Ina Otte et al., 2017. Switzerland GP's-20	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
19. Ada van de Scheur, Arie van der Arend, 1998 Netherlands Nurse-20	Y	Y	Y	Y	Y	N	N	Y	Unc	Y	include	
20.Belanger E.et al., 2019 Canada-palliative care physicians-18	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
21. Jessica Shaw et al., 2018. canada phy-8	Y	Y	N	Y	Y	N	N	Y	Unc	Y	Include	
22. Judith Schwartz 2004. USA nurses-10	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	

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23. Dobscha SJ et al., 2004. USA phy-35	Y	N	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	No theme of emotional impact.
24. Galusko et al., 2015, Germany 19 specialized palliative care physicians.	Y	Y	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	Desire to hasten death-definition ambiguous
25. Susanne Brauer et al., 2015. Switzerland, 12 physicians	Y	Y	Y	Y	Y	N	N	N	N	Unclear	**Exclude**	Opinions known, but no emotional ipact theme
26. Linda (b) Oregon phy-35	N	Y	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	Physician opinion of patients req
27.Deborah-texas nurses-36	N	Y	N	Y	Y	N	N	Y	Unc	Y	**Exclude**	No of the nurses participated in assisted suicide in any way
28. D Van Rooyan, Dutch nurses-7	N	N	Y	Y	Y	N	N	Y	N	Y	**Exclude**	More with withdrawal of treatment does not meet criteria
29. vanderspank canada Nurses	N	N	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	SR on nurses experience with withdrawal of treatment-does not meet criteria
30. Joanne Wolfe USA 324 Oncologists	Y	N	N	Y	Y	N	N	N	Y	Y	**Exclude**	Telephone based survey interviews.
31. Booij et al., 2012 Netherlands 15 physicians	Y	N	Y	Y	Y	N	N	N	Y	Y	**Exclude**	No particular description of emotional impact
32. Denier et al., 2010 Belgium 18 Nurses	Y	N	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	More about communicati

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												on and communicati on attitudes and not about emotional impact
33. Bernadette Dierckx 2010 Belgium 18 nurses	Y	N	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	Stage of carrying out a request, no emotional impact described.
34. sercu et al. 2012	Y	N	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	Palliative sedation and euthanasia-boundry lines unclear in the paper.
35. Volker 2007 USA. 19 oncology advanced practice nurses	Y	N	Y	Y	N	N	N	Y	Y	Y	**Exclude**	No engagement in assisted death as illegal in the place of practice.
36. Thulesius et al. 2013 Sweden	Y	N	Y	Y	N	N	Y	N	Y	Y	**Exclude**	No engagement, assisted death is illegal in Sweden. Majority data from HCPs in Sweden.
37. Marike E. de Boer 2011 Netherlands.	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	Experiences, but no emotional impact
38. Neel De Bal 2006 Belgium	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	Conducted at a time when Euthanasia was still illegal, hence does not meet

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												inclusion criteria.
39. Bernadette 2006 Belgium	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	As above.
40. Veerport et al 2006 USA	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	As above
41. Wright et al., 2017 Canada	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	Data collected in 2012-2013 when MAiD illegal.
42. Curry et al., 2000 USA, Connecticut 909 physicians.	Y	N	Y	Y	N	N	N	N	Y	Y	**Exclude**	Assisted suicide illegal, Plus experiences and no emotional impact
43. Susan Price 2001 USA, 11 nurses and 10 physicians. North Carolina	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	Assisted suicide illegal in North Carolina, hence does not meet inclusion criteria
44. France Norwood 2009 Netherlands	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	No emotional impact. Evaluates absence of abuse
45. Smith et al., 2013 USA, South Mississippi	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	Assisted death illegal in mississippi and hence does not meet inclusion criteria
46. Beuthin et al., 2020 Canada 8 physicians.	Y	Y	Y	Y	N	N	Y	Y	Y	Y	include	
47. Khosnood et al., 2018 19 physicians, Canada	Y	Y	Y	Y	N	N	Y	Y	Y	Y	Include	

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48. Pesut et al., 2020 59 RN and NPs, Canada	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	include	
49. Keri-Lyn Durant and Katherine kortes Miller 2020 Canada	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	Include	
50. Snijdewind et al., 2016 Netherlands 28 physicians	Y	Y	Y	Y	N	N	Y	Y	Y	Y	include	
51. Mathews et al., 2021. Canada 23 palliative care providers (13 physicians, 10 nurses)	Y	y	Y	y	N	N	Y	Y	Y	Y	include	

Total Included studies (N_1): 27

Databases: OVID Medline, EMBASE, CINAHL, SCOPUS

Critical Appraisal tool: The Joanna Briggs Institute Critical appraisal tools for use in Systematic reviews: checklist for Qualitative research Available from:
http://joannabriggs.org/research/critical-appraisal-tools.html

Discussion: Lockwood C, Munn Z, Porritt K. Qualitative research synthesis: methodological guidance for systematic reviewers utilizing meta-aggregation. Int J Evid Based Healthc. 2015;13(3):179–187.

Grey Literature databases (December 10th 2018 to March 1st, 2019, updated August 2020)

Total # Records after de-duplication: 13. Records selected after applying critical appraisal tool 8.

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Database	Search strategy	#records screened	# new records selected after applying after de-duplication and applying critical appraisal tool
Google scholar	With the exact phrase: “Medical assistance in dying” ; “physician assisted suicide”; With all the words: “emotional impact on health care providers involved in medical assistance in dying”	400	5
Des Lebris/Canadian Electronic Library	Medical assistance in dying	5	0
Canadian Institute of Health Information (CIHI)	Medical assistance in dying	7	0
OAIster database (includes WordCAT)	Medical Assistance in dying, Physician assisted suicide as key word	206	2
OpenGrey	Medical assistance in dying, Physician Assisted suicide as key word	4	0
BASE (Bielefeld Academic Search Engine)	Subject Heading search: “Medical Assistance in dying”	670	1

Selected records:

Google scholar included Results:

1. Beuthin R. Cultivating Compassion: The Practice Experience of a Medical Assistance in Dying Coordinator in Canada. *Qualitative Health Research*. 2018;28(11):1679-1691. doi:10.1177/1049732318788850

2. Bouthillier M-E, Opatrny L. A qualitative study of physicians’ conscientious objections to medical aid in dying. *Palliative Medicine*. 2019;33(9):1212-1220. doi:10.1177/0269216319861921

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3. Bruce A, Beuthin R. Medically Assisted Dying in Canada: "Beautiful Death" Is Transforming Nurses' Experiences of Suffering. *Canadian Journal of Nursing Research*. June 2019. doi:10.1177/0844562119856234

4. Alison A. Townsley. Putting a Bow on Death and Dying-Health Care Professionals' Experiences with Medical Assistance in Dying (MAiD): A Foucauldian Discourse Analysis with Agambian Insights. [Practice Based Research Paper on the Internet]. York University School of Social Work; 2018. [Cited February 28,2019] Available from: https://yorkspace.library.yorku.ca/xmlui/bitstream/handle/10315/35613/Alison_Townsley_PRP_2018.pdf?sequence=1

5. Gamondi C, Borasio GD, Oliver P, Preston N, Payne S. Responses to assisted suicide requests: an interview study with Swiss palliative care physicians. BMJ Support Palliat Care. 2019 Mar;9(1):e7. doi: 10.1136/bmjspcare-2016-001291. Epub 2017 Aug 11. Available from https://pubmed.ncbi.nlm.nih.gov/28801317/

OAIster included Results:

1. Implementing Medical Aid-in-Dying in Vermont: a Qualitative Study. *J GEN INTERN MED* **34,** 636–641 (2019). https://doi.org/10.1007/s11606-018-4811-1

2. Allyson Oliphant. "If Not Me, Then Who?" The Narratives of Medical Aid in Dying (MAiD) Providers and Supporters Around Their Professional Identity and Role in MAiD [Thesis on the Internet]. MacMaster University; 2017. [Cited on February 10, 2019]. Available from http://hdl.handle.net/11375/22146

BASE included results:

1. Sheridan, Laura, "When Patients Ask to Die: The Role of Nurses in Medical Assistance in Dying" (2017). Electronic Thesis and Dissertation Repository. 5041. https://ir.lib.uwo.ca/etd/5041

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Table 2: Description of articles included in qualitative meta-synthesis:

Study	Number and country of origin of participants	Description of participants	Extent of engagement in the MAiD process	Method of interview	Method of analysis	Emotional theme explored
1. Voorhees et al., 2014	23 physicians, 18 from USA (5 from Oregon), and 18 from Netherlands	@40% from primary care, majority >40 years	Physician assisted dying discussions.	40-70 min, one-one semi structured interviews	Modified 5-step framework-familiarization, identifying a theme, indexing, charting, mapping and interpretation.	Themes related to reflective emotions and sense of growth along with themes emotional labor and conscientious-based emotions.
2. Marwijk et al.,2007	22 primary care physicians, Netherlands	Variable range of experience, 5 PCPs participated in the Support and Consultation Regarding Euthanasia (SCRN)	Discussing and performing assisted death	4 focused groups, homogenized as per age and gender.	Content analysis within a coding frame of three themes of (1) emotional experience; (2) coping (dealing with and managing the event) and (3) role of the physician.	Themes related to reflective emotions and sense of growth along with themes emotional labor and conscientious-based emotions.
3. Denier et al., 2010	18 nurses from 5 provinces of Flanders, Belgium	Registered nurses (13 women, 5 men) of geriatric, oncology, internal medicine, and palliative care. All had positive attitude, except one who was conscientiously objecting.	Discussing and performing assisted death	1.5h in-depth interviews, think back to a specific, recent case of caring for a patient requesting euthanasia and to recount the way in which they experienced this	Grounded theory design	Themes related to role-assigned emotions along with themes of emotional labor.

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				process as a whole		
4. Norton et al., 2012	9 social worker hospice practitioners in Oregon, USA.	Represent several health systems in Oregon	involved in discussions with family of those participating in assisted death ('add on') and 'context interpreters'	Focused group	Thematic analysis	Themes related to role-assigned emotions (for example advocacy and feeling of being a 'gate-keeper')
5. Georges et al, 2008	30 general physicians in Netherlands.	71% male, 29% female, 46% had restrictive and 14% had permissive attitudes towards euthanasia.	89% had received explicit requests and were involved in discussions, and 64% had participated in EAS	In-depth interviews	Constant comparative method of analysis	Emotional theme of reflective emotions (example, feeling of sense of growth)
6.Snijde wind et al., 2014	28 General Physicians in Netherlands	Physicians who had received a request from someone suffering from dementia or a psychiatric illness, or who was ''tired of living,'' as these are cases that are often regarded as complex.	Involved in decision making of assisted death for respective patients.	In-depth interviews	Open coding and inductive analysis	Emotional theme of reflective emotions (example, reflecting on individual meaning of suffering)
7. Katja ten Cate et al., 2017	15 General Practitioners in Netherlands	8 GPs with liberal attitude, 5 with conservative attitude and 2 with neutral attitude towards assisted death. Mean age 51.2 years.	1-2/>2 assisted deaths performed.	In-depth interviews	several phases of coding (axial and selective coding); codes were refined, sub codes and overarching codes were assigned and relationships between codes were explored. Interviews were also analysed as a whole, to look for	Emotional theme of reflective emotions (example, reflecting on feelings of what is happening during the last stage of life)

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					patterns and inconsistencies in reasoning.	
8. Donald G Van Tol et al., 2012	15 physicians in Netherlands	Fourteen of them were general practitioners. Seven of them were also active as a consulting doctor, one was a nursing home doctor who was also working as a consulting doctor.	Physicians were consulting doctors of Euthanasia and have successfully completed a formal training program.	In-depth semi-structured interviews	Grounded theory approach by Glaser and Strauss and Glaser	Emotional theme of reflective emotions (example 'imagine self', cognitive reflection)
9. Melchor Lorraine 2018	8 social workers in California, USA.	75% female with 60% having an average 5 years of experience in hospice care.	assist patients and family with the death and dying process, may connect them to additional community resources, and offer counseling to improve and maintain emotional, psychological, social, and physical well-being	In-depth semi-structured interviews	Open coding, axial coding, selective coding, and conditional matrix stages of data analysis.	Emotional theme of role-assigned emotions (example, feeling of pro-self-determination and advocacy).
10. Miller et al., 2002	8 social workers in Oregon, USA	2 men, 6 women, age range of 27-64, 3-22 years' experience in hospice care	Active engagement in end-of-life care and assisted suicide discussions.	interviews	Ethnographic study and constant comparative method of analysis	Emotional theme of role-assigned emotions (example advocacy and self-determination)
11. Beuthin et al., 2018	17 Nurses in Canada	NPs, RNs, and LPNs, from urban and rural areas across Vancouver Island, British Columbia, working across	15 nurses had direct experience with MAiD, 7 were involved in some aspect of assisted death in the patient's journey (e.g., providing	In-depth semi structured interviews	Descriptive narrative enquiry and thematic analysis	Emotional theme of reflective emotions (example, a sense-making process)

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		settings including acute care, residential care, primary care clinics, and community and palliative care.	information, acting as witness to the medical assessment, providing care before or after, etc.)			
12. Bolt et al., 2016	8 pediatricians in Netherlands	8 pediatricians who were interviewed were 5 men and 3 women, aged 44–62y, working in four academic and three general hospitals .	25% had received an explicit request for Physician-assisted death, with 7% in the last two years, and the requests were mostly made by parents (25%) and sometimes by patients (6%)	Semi-structured interviews	Qualitative Analysis Guide of Leuven method was used for the analysis. Mixed method approach.	Emotional theme of role-assigned emotions (example, feeling of duty)
13. Dolares Angela Castelli Dransart et al., 2017.	1 physician, 8 directors of sociomedical institutions or organizations, 10 head nurses, 8 nurses, 10 nursing assistants or care assistants, and 3 sociocultural animators, Switzerland confronted with assisted suicide requests.	27 men, 13 women, mean age 52y.	14 had been faced with suicide or assisted suicide in their personal life, beside the situation of assisted suicide at work. None of the respondents interviewed had physically provided the lethal substance to perform the assisted suicide (a task assigned to Right to Die associations), nor were they directly involved in the decision-making process that enabled the assisted suicide to take place (except for one physician). In fact, the vast majority of these professionals	Semi-directive interviews conducted at workplace.	Grounded theory using 3 types of coding-open, axial and selective.	Emotional theme of role assigned emotions (example, feeling of professional compromise)

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			(except for two) declared that not only did they appreciate the fact that Right to Die associations assumed the task of delivering the lethal substance and physically assisting the requestor, but they also did not want to be led to do it themselves in the future			
14. Marianne Dees et al., 2012	28 physicians in Netherlands	20 males, 8 females, 22 GPs, 1 elderly care 2 GP trainees and 1 psychiatry	once in 3-5 years' experience with assisted death.	In-depth interviews with patients who had explicitly requested assisted death, their most involved relatives and their treating physicians	Thematic analysis	Emotional theme of reflective emotions (example, relational and feeling of trust in physician-patient relationship)
15. Harvath et al., 2006	20 hospice social workers and nurses in Oregon, USA.	--	The 20 hospice social workers/nurses described 33 different cases of terminally ill patients who had requested them to hasten death through physician assisted suicide (n = 22)	Semi-structured, In-depth interviews.	Thematic analysis	Emotional them of role-assigned emotions (example, feeling of professional failure, professional dilemmas and inner debate).

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16. Ina Otte et al., 2016	20 General practitioners (GPs) in Switzerland., 3 declined to participate due to personal discomfort with assisted death.	GPs who had chosen to refuse to assist a patient's suicide comprise the largest group in the study and provided the most insights.	Receive 1-3 requests of physician assisted suicide per year. 2/3rd of the GPs interviewed had chosen to refuse to assist a patient's suicide comprised the largest group in the study and provided the most insight into their handling of requests for PAS.	In-depth semi-structured interviews.	Thematic analysis	Emotional theme of basic emotions with conscience-based avoidance/rejection of MAiD (example, feeling of moral distress)
17. Ada van de Scheur and Arie van der Arend 1998	20 nurses in Netherlands	According to different phases of Euthanasia: Observation of a request for euthanasia: 17 nurses. 2) Decision making: 14 nurses. 3) Carrying out of euthanasia: 12 nurses. 4) Aftercare: 14 nurses	Engagement as per different phases of Euthanasia	In-depth semi-structured interviews.	Thematic analysis	Emotional theme of role-assigned emotions (example, feeling of moral distress)
18. Emmanuelle Bélanger et al., 2018	18 university affiliated palliative care physicians in Quebec, Canada	Participants positioned themselves opposite euthanasia	majority of the palliative care physicians on staff at the palliative care units of two public hospitals located in an urban area of Quebec. All participants were full-time palliative care physicians, and like most palliative care providers in	In-depth semi-structured interviews.	Inductive methodology of Interpretive description.	Emotional theme of role-assigned emotions (example, professional dilemmas and conflicting values with palliative care)

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			Canada, the majority of them (16 out of 18) were family physicians. As expected, all participants expressed discomfort with euthanasia as an aspect of end-of-life care. All but one denied the influence of religious or political positions in shaping their views.			
19. Jessica Shaw et al., 2018	Eight physicians who offered MAID in British Columbia in 2016, Canada	3 were from greater Vancouver, 3 were from Victoria, and 2 worked in a small community on Vancouver Island. Seven were family doctors and 1 was a general internist. Their ages ranged from 37 to 64 years. There were 2 men and 6 women; 6 worked full-time and 2 worked part-time.	Collectively, by the end of December 2016, the 8 physicians in this study had assessed 332 people who were seeking MAID and had completed 135 assisted deaths	In-depth semi structured interview via phone call/email	Qualitative thematic analysis	Emotional theme of basic emotions, especially positive emotions (example, sense of fulfilment)
20. Judith Schwarz, 2004	10 nurses who worked in home hospice, critical care, and HIV/AIDS care settings, USA	Four worked in hospice home care, three were advance practice nurses who worked with persons with AIDS, two worked	Nurses were eligible to participate in this study if they believed that a competent patient had made a serious request for their help in dying.	In-depth interviews done at least twice for 7 participants .	van Manen's approach to phenomenology phenomenological interpretation and analysis (phenomenological enquiry)	Emotional theme of role-assigned emotions (example, feeling of human-human response and connectedness)

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		in critical care, and one was a clinical nurse specialist in the care of patients with spinal cord injuries. Two of the ten nurses were male, all were Caucasian, middle-aged, well educated (three PhDs; five Masters of Science in Nursing), and clinically experienced (6–35 years)				
21. Marie-Eve Bouthillier and Lucie Opatrny 2019	22 conscientiously objecting physicians in Quebec, Canada	26 to 67 years (mean: 45 years), 12 of them were male (54.5%). 14 Family physicians, 2 oncology and 1 each from psychiatry, neurology, nephrology, intensive care, geriatrics and pneumology. 14 from catholic background.	Physicians had received requests, had discussions with patients regards to MAiD, and conscientiously objected to participate.	Semi-structured interviews. eight open-ended questions Interviews ranged in length from 15 min to 1 h, with a mean length of 24 min (median length = 21 min). think back to their first medical aid in dying request (as some physicians had received	descriptive thematic analysis	Emotional theme of basic emotions (for example emotional labor, burden and fear of psychological repercussions)

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				more than one request) and describe the reasons which motivated their refusal.		
22. Gamondi et al., 2017	23 palliative care physicians across Switzerland	65% German, 30% French and 5% Italian speaking	Regularly received assisted suicide requests. The involvement of Swiss physicians is mostly confined to the decision-making phase; medical certification of diagnosis and mental capacity.	Semi-structured interviews.	thematic analysis	Emotional theme of role-assigned emotions (example professional role-related feeling of ambiguity, fear of being stigmatized as physicians, feeling of walking a tight rope.)
23. Rosanne Beuthin, 2018	female, of Anglo-European ancestry, age mid-fifties, living in an urban center, Canada	Doctorate in nursing and was employed as a consultant under an end-of-life Program to enact a new MAiD program.	daily journal entries made over a 6 month period, from the first day of immersion in the role and culture of MAiD from late May to October 2016	Raw autobiographical text held scattered floods of ideas and released emotions into a thick created Story.	autoethnographic approach-reflective analysis	Emotional theme of reflective emotions (example, feeling of embodiment, compassionate care and sense-making reflective emotions. Exploring tensions around language, attitudes)
24. Anne Bruce and Rosanne Beuthin, 2019	15 RNs/NPs/LPNs from British Columbia, Canada.	Participants worked in diverse settings including acute care, community-home care, and specialty areas including emergency	Eight nurses had directly aided with MAiD and cared for the patient at home or in a care setting. Seven had been involved indirectly with patients such as providing assisted	Semi-structured interviews- (1) tell me about your first experience of being asked to participate	narrative inquiry and thematic analysis	Emotional theme of reflective emotions (example fear of desensitization with deeper questioning) along with complex emotions of "compassion

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		room and palliative care.	dying information upon request and listening to patients and families as they explored pursuing MAiD	in a medically assisted death and how you came to the decision to participate or not and (2) tell me about the MAiD experience itself. What was most challenging ?		satisfaction" as well as compassion fatigue
25. Alison Townsley 2018	seven nurses, social workers, and personal support workers, Canada	Health care professional enrolled through purposive sampling.	Engaged in discussions and assessments of patients requesting MAiD.	one-on-one, semi-structured interviews with health care professionals	Foucauldian Discourse Analysis perspective. Interview data is analyzed by situating the health care professional as an effect, as a producer, and as a challenger of power-knowledge systems. Philosophical theories of Giorgio Agamben are applied to the data to challenge Foucauldian principles, and to bolster the discussion of defining of the body that deserves to live,	Emotional theme of reflective emotions (example, emotions emerging from engagement of the individual in terms of power, knowledge and individual identity)

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					and the body that deserves to die.	
26. Buchbinder et al., 2019	37 health care providers in Vermont, USA.	Health care providers from Hospital and community-based practices. Most were women (68%) and the largest subgroup specialized in internal or family medicine (53%). Most of the nurses and social workers were women (89%) and most worked for hospice and home health agencies (61%).	19 physicians (10 internal medicine, 4 palliative care, 3 neurology, 2 oncology), 12 had participated in Act 39 (The patient Choice and control at End-of-Life Act) as prescribing physicians, the remainder had initiated but not completed the Act 39 protocol (n = 3), participated as a second physician to confirm the patient's diagnosis, prognosis, and decisional capacity (n = 3), or counseled patients (n = 1). The mean age of nurses and social workers (n=18, 9 hospice/home nurse, nurse practitioner 5, inpatient palliative care 2, hospice social worker 2) was 52.5, with most working for hospice and home health agencies (61%). While all professionals in this group engaged in clinical care for patients pursuing Act 39, specialty clinic nurse	One-to-one semi structured interviews	Grounded theory approach	Emotional theme of role-assigned emotions (example pride, burden etc.)

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			practitioners were more likely to assist with navigating access to the aid in dying. Participating health care professionals worked in ten of Vermont's 14 counties			
27. Allyson Oliphant, 2017	4 physicians. 4 nurses and 6 HCPs (allied health care professional social workers (1), spiritual care providers (1), pharmacists (1), genetic technologists (1) and psychologists (2).) of team ADRAS in Hamilton, ON.	Of the data available, 2 were semi-retired family physicians, One is an intensive care physician with a background in cardiology, and the second is an Emergency Room physician with training in palliative care.	All participants are members of the ADRAS (assisted dying resource and assessment service) who support the practice of MAiD. Every participant had a capacity to be flexible.	One to one semi-structured interviews.	Grounded theory approach	Emotional theme of reflective emotions (example, emotions related to related to professional identity, sense making, feeling of obligation to serve)
28. Laura Sheridon 2017	nine palliative care nurses in southwestern Ontario, Canada	3 males, 6 females. 3 participants worked in residential hospices where MAiD was not supported as an end-of-life option, six participants worked in the community providing home care where MAiD is an option in end-of-life planning. Two participants had	Participants in the study indicated that nurses may act as a liaison between physicians and nurse practitioners who have the authority to assess patient eligibility and provide the intervention of MAiD and the patient, notifying them of an inquiry about or a request for MAiD	One-to-one semi structured interview.	interpretive description qualitative methodology	Emotional theme related to role-assigned emotions (example, emotional expressions ("hard conversations") related to nursing role, struggle related to moral conflicts.

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		previous inpatient hospital experience in emergency care and in intensive care specialties.				
29. Khosnood et al., 2018	19 physicians, Canada. Quebec not included.	Half of the participants were palliative care specialists (n = 8), with the remaining representing Family Medicine (n = 4), Anesthesia (n = 2), Hematology (n = 1), and Obstetrics & Gynecology (n = 1). The majority of participants practiced in an urban setting (n = 13).	Average 6.9 MAiD cases.	In-depth semi-structured telephone-based interviews.	inductive thematic analysis approach	Emotional theme of role-assigned emotions (example burn out, negative effect on inter-professional relationships vs. increased feeling of respect)
30. Beuthin et al., 2020	8 physicians, Canada.	Participants included general practitioners (GPs) and Non-specialist physicians from urban and rural communities working in acute and palliative care. Ages ranged from 33 to 62 years (average age 49), with an equal number of men and women. The majority identified no active religious	experience with MAiD provision ranged from 12 to 113 assisted deaths. Only one physician was dedicated to full-time provision.	In-person or telephone-based semi-structured interviews.	interpretive descriptive methodology and thematic analysis	Emotional them of reflective emotions, (example complex emotions of compassion satisfaction, embodied awareness, soul-searching)

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		affiliation, and ethnicity was withheld to protect anonymity. Years of experience ranged from 6 to 38 years (average of 23).				
31. Keri-Lyn Durant and Katherine Kortes-Miller 2020	23 physicians of Rural area, northwestern Ontario, most of subarctic Ontario.	23 physician participants ranged in age from 26 to 63, with a mean age of 43 years. Physicians worked in a variety of settings, with 14 in an urban setting – in family practice, as a hospitalist or other specialist, in the emergency department, in palliative care, and in long-term care. Nine participants declared a rural practice, and self-identified as rural generalists, working on a First Nations' reserve, in a community, at a satellite clinic, or 'All of the above'.	11 identifying themselves as acting both as assessor and provider, 1 as assessor only, 4 as providing referrals upon request, and 7 without any direct/indirect experience. These seven were included in the study because they expressed a desire to participate and reported that their practice and the community had been impacted by the legislation. There was also a variance in terms of exposure to death in practice, with an estimated total between 2 and 250 deaths per annum	using 1 semi-structured focus group and 18 semi-structured interviews comprising 9 set of questions	Thematic analysis	Emotional theme of role-assigned emotions (example, feeling of impact on inter-professional relationships, feeling of unpreparedness.
32. Snijdewind et al., 2016	secondary analysis of in-depth	Respondents were recruited both by the network of physicians	Twenty-two respondents worked as family physicians, and six	One-to-one semi-structured interviews.	Thematic analysis	Emotional theme of reflective emotions (example, those related to meaning

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	interviews with 28 Dutch physicians who had experience with a complex case of EAS	working for SCEN (Support and Consultation for Euthanasia in the Netherlands) as well as via a national Questionnaire. Nine of the respondents were female. The respondents' age ranged from 36 to 68 years	worked as medical specialists (three elderly care physicians, a psychiatrist, an internist and a lung specialist). Next to this, six of the respondents also worked as SCEN physicians. All had experience with EAS requests and the performance of EAS.			of suffering, blurring emotional boundaries)
33. Pesut et al., 2020	59 registered nurses and nurse practitioners in Canada	n = 9 (15%) were conscientious objectors, Spiritual or Religious Affiliation: n = 33 (56%) Neither: n = 15 (25%); Spiritual but not Religious: n = 11 (19%) Home & Community: n = 32 (54%); Acute Care: n = 10 (17%); Long-term care: n = 5 (9%); Hospice: n = 4 (7%); Clinic: n = 3 (5%	24 of the 59 participants had conducted more than 25 conversations with patients about MAiD, and 11 of the 59 participants had been involved with more than 25 patients who went on to receive MAiD.	Semi-structured interviews conducted on telephone. Question examples: (i) Can you tell us how the process of MAiD occurs in your practice context? (ii) What resources and practice supports are available to assist you in caring for MAiD patients? (iii) Tell us about your experiences with MAiD?	Qualitative approach guided by Interpretive Description. data immersion, open coding, constant comparative analysis, and the construction of a thematic and interpretive account. Transcripts include emotions evident during the interview (e.g., crying).	Emotional theme of role-assigned emotions (example, emotions related to find themselves caught between the proverbial "rock and hard place." With feelings of Emotions of frustration, powerfulness of the experience, feeling drained out)

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				The average length of interviews was 55 min.		
34. Deborah Volkar et al., 2001	40 oncology nurses who received requests for assisted death in USA.	48% in hospital/multi-hospital settings. 9 female, 1 male. Mean age 45 y.	30% had received requests for assisted suicide, 6 (1%) engaged in assisted suicide, and 20 (4.5%) admitted to intentionally injecting a drug to end a patient's life.	Recipients were requested to submit a written account or story of receiving a request for assistance in dying from a terminally ill patient with cancer.	Denzin's process of interpretive interactionism with an emic, ideographic approach. That is, individual experience is considered to be unique; discovery of an individual's epiphany and associated meanings is the research focus	Emotional theme of basic emotions (example emotional labor) along with reflective emotions of feeling lack of control (or lack of it) and moral distress).
35. Mathews et al., 2021	23 palliative care providers (13 physicians and 10 nurses) who practiced for 6 months or more before and after the introduction of MAiD, in inpatient and community-based settings that supported assisted death in southern Ontario, Canada.	54% of physicians and 90% of nurses were female with a mean age of 43 years and 42.6 years respectively.	All the participants described having discussions with patients regarding MAiD and 7/23 participants (4 nurses and 3 physicians) described directly witnessing assisted death. 8/13 physicians made referrals for MAiD, 4 conducted assessments, and 3 physicians were MAiD providers; 3 physicians identified as conscientious objectors. None of the nurses identified themselves as conscientious	Semi-structured interview based on pre-determined interview guide	Braun and Clarke's version of Thematic analysis	Emotional theme of role-assigned emotions (example Role-driven emotional themes of Emotional, psychological and resource burden along with theme of emotional labor)

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			objectors, although some expressed moral or religious conflict around MAiD.			

Table 3: Codes and Themes table (corresponding Table-2 study number in parenthesis):

A) Over-arching theme of basic emotions:

Theme 1: Emotional labor (positive/negative emotions) Codes:	Theme 2: Conscience based emotions. codes:
"rewarding" "liberating", "Well please let someone else do this question", "blood had frozen in my veins", I just felt just totally cold all over. I had no idea of what to do. I realized there was no help I could get from anywhere . . . I . . . felt as though I was . . .impotent to help them. "If possible, I would run away. But I see it as the last part of my care. I have taken care of that patient for years and now at the moment . . . when she needs me most . . . I would be a coward to run away then. (1) "I felt very lonely" "heroic feelings", "tense", "scary", "terribly creepy", "felt pressured to succeed", "suffer a loss yourself when someone like that dies" "terribly manipulated", "felt slightly put upon, angry" 'let off steam' (2) "feeling of ambivalence", "intense", "gradually feel less secure, less fearful", "surprisingly grateful" . "very demanding and emotionally distressing" (3) "very demanding, generally like to avoid", "drastic"(5) , "moral pressure" , "uncertain, complex"(6), "very hard"(7), "feeling choked up or shedding a tear" "Feeling positive emotions of peace and amazement were more surprising and often shared cautiously in public" ,"had difficulty finding effective words for the paradoxical experience of witnessing death that is, both "sad" and "beautiful." (11). "felt reluctant as it is difficult to predict" (12). "feeling of enrichment", "feeling of sorrow and intrusive thoughts", "feeling like weathering the storm", "empathy and emotional closeness", "personal compromise" (13). "do not feel competent" (16).	"making pluses and minuses about it . . . but . . . 'What's it doing to me? I'm going to kill someone tonight.'[respondent began to cry], "I have to do no harm, and I just feel that if you're assisting someone in dying . . . it's against what I've been trained . . . It's not up to me to decide when the patient dies . . ." (1); "killing another person is not the solution. It's in the ten commandments" "sense of guilt. I feel as if I'm an executioner. Who am I to have the right to do this?" (2); "Conscientiously, I find it hard to come to terms with euthanasia" (3); Clarity of conscience- "a sort of trap that can't be avoided. That in spite of everything you can offer, a terminal stage can be so heavy, perhaps too heavy for a patient. In fact, I always see it as an emergency exit. When I am talking about it with a patient I say, "yes we will consider it, if you don't want to go on any longer and if I have nothing more to offer you to make it better"(5); "I am a Christian so I have strong feelings because of my belief and my background, believe that no human being should be in the position to hasten death." (10); cannot bear the idea of killing one of my patients", I do not feel competent to deal with the topic…especially for my personal psychological health, "challenges my belief, I do not understand how it can be meaningful" (16)

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"rewarding work", "honor", "bit overwhelming", "proud", "incredible" "feeling like being on call all the time (19), "emotional burden", "fear of psychological repercussions", "uncomfortable", fear of stigmatization (21), "fear of stigma/isolation, feeling of ambiguity" (22), "feeling courageous" (23), "satisfying and gratifying" " roller coaster" , "transformational feelings of beautiful death" (24), just feel coldness, or whatever. You just feel drained …"(28), "unexpected rewards" , "enriching capacity of caring", (30), , "anxiety, shock, self-doubt", "deep inside…conflict" (34); "walking quiet a tight rope", was as prepared…but went outside and felt like I was about to throw up", "actually, find them. . . they're such beautiful experiences with family. It's the shared experience with the family that you're with that you have an opportunity to help." (35)	"to see somebody lying there, to whom you brought a cup of tea that morning. And you know that everybody who gets a heart attack can die as well, but this was no heart attack. You know that, of course. So, somebody has been killed, just like that. . . That makes it different." (17) "conflicted, trying to reconcile their own personal moral stance with facilitating the end of someone's life" (28) "What would my family think that I'm working on a unit that does that [Medical Assistance in Dying]? Do I hide it from them. . .what if people find out that we do it? Are people going to come up here and start protesting? People will see that as evil." (35)

B) Overarching theme of reflective emotions.

Theme 1: relational "feeling of trust and sympathy in physician patient relationship strong" (14) "human centered, compassionate care" (23), "for somebody to approach you is almost an honor that they trust you enough to have this conversation, and to have to sort of shut them down, or acknowledge how they're feeling" (empathy) (28), "intimate, emotional engagement-rediscovering the art of medicine", (30), "indelible nature of the experience shared" (34) "as soon the topic [Medical Assistance in Dying] came up, that I was a conscientious objector and the person said that you're not on my side, even though she was getting the service [MAID] . . .I was seen as somebody who was not helping her" (35)	Theme 2: Discourse based (control over a natural process of dying) "interesting discourse presented itself through idea of using stages to determine someone's chances of survival, and the need for professionals to have something finite and concrete to measure", "discourse that emerged through conversations with participants was how control (or masterhood) equates to people's sense of wellbeing" "MAiD itself presents a paradox insofar as one can be too sick to access this form of assistance that is exclusively designed to bring death to the most critically ill people" "The most dominant discourse that emerged from this data set was participants aligning what is right and good within the confines of the law." (25); "medicalization of a social problem" (32); "degree of control over dying process" (34).
Theme 3: Sense making process and related emotions. (Theme of Growth) "You grow with the problems of the patients" (1) " stay closer to their own beliefs" " long road to becoming aware of one's own views" (2) "meaning full experience" " almost closer than when someone is having a baby" (5) "[EAS] is not an act, it's a process towards which we both grow" (6) , "Being in process, holding an in-between space of uncertainty, reflection, and active sense-making" (11); "pure moment of autonomous self-consciousness" "I am working and sense making as I go along, being sure that I keep breathing",	Theme 4: Process influenced themes (suffering---relief--death) "Invisible suffering made it harder for the people close by to empathize and come to terms with the patient's request and his/her death" (6); "for me, a lot of talk, talk about death and dying, talk about life, about saying goodbye, really seeing and feeling what is happening in this last phase of life and reflect on that. But not everybody is capable of talking and reflecting this way, while everybody is going to die. So that's my problem" (7); "imagine self" and "imagine other" cognitive route. Use of cognitive reflection (8);

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"feeling of embodiment, become the face of MAiD", "bearing witness"(23); "worries of becoming desensitized and ongoing deeper questioning" (24); "their thoughtful silence after speaking or listening represented and solicited from me respect for the dead and the dying, seething inner anger, and perhaps the quietude that one experiences when their physical body feels the effects of being a challenger and resister in the strongest way possible" "Kind of letting them have control over what they can have control over" "beautiful journey of self-reflection", "grappling with identity" (25); "embodied awareness", "soul searching" (30); "silent knowing" (34)	"very difficult for me to let…go, to be so aware of saying farewell, and now I notice that as time passes it gets harder and harder for me" (14); "sense of urgency to hasten death" (23); "boundaries of EAS has shifted over time, making feel stretched, tense and insecure" "not feeling competent if suffering is existential" (32); "it's been a bit of a challenge to delineate what we're doing in relationship to the request for assisted dying and what normal care still continues to be" "struggle with the rules of a complex legislated and reporting process that determines it"(33)

C) Overarching theme of emotions related to professional values:

Theme 4: Role-assigned emotions **Nurses:** "predominantly tend to be conformist (following existing conventions rather than using critical reflection) when faced with ethical dilemmas. Combined with the emphasis of the medical responsibility in euthanasia care, and combined with the strong inclination of nurses to respect the patients' wishes, it seems logical that nurses interpret the gravity of the process in emotional terms"(3); ""unchartered territory," where "there was almost no foundation" for providing this option, and "this is a whole new role for all of us.(being pioneers)" "duty to provide care" is being touted as "you don't have a choice" and the information isn't there [about] how to object if you don't agree with" (11); "moral distress", "burden", see somebody lying there, to whom you brought a cup of tea that morning. And you know that everybody who gets a heart attack can die as well, but this was no heart attack. You know that, of course. So, somebody has been killed, just like that. . . That makes it different" (16); "identifying the moral line", "human-human response and connectedness because of the role played", "fear the potential for abuse, and the possibility that other health-care professionals might too readily accept a patient's fleeting wish to die" (20); "taken for granted, feeling terrible" "their own suffering is invisible" (24); "walking alongside patients" like the experience of being able to make [death] a better experience. That celebration of life rather than the mourning of death" (27); "feeling of having hard conversations" (28); "Nurses seeking to provide the compassionate care consistent with such a momentous moment in patients' lives, without suitable supports, find themselves caught between the proverbial rock and hard place" "powerful experience" "mad as a hell", "overwhelmed" "…don't find the provisions so emotionally draining, but it's more the logistics and it's a lot of work as a nurse" (33); there's a sense of ceremony [before Medical Assistance in Dying], So, those all have impacts in terms of resources" (35). **social worker**: "feeling of being a gatekeeper" (4); "sense of preparedness", feeling that this option is 'pro-self-determination which is our job"(9); "inner debate, cannot make peace with that, felt a huge shift in my ethics", "dying process has a lot to give" "missed opportunity to deepen oneself spiritually", "missed opportunity to forgive"(15); feeling of advocacy and self-determination in sync with hospice and social work values, and we will advocate for the patients . . . to get them whatever they want . . . I believe in self-determination, but I think it's (PAS) a sad commentary on our society." "Our job is to meet the patients where they are" (10); "felt like higher commitment", "felt like a failure if patient chose EAS" (16). **physicians**: "heavy responsibility" (5); "implicit ethical tension due to pressure to decide", "It is the right time for EAS] Only if someone is totally at peace with himself, his life and his death, and if I see and feel that too.'(7); "feeling of duty" (12);

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"professional compromise" (13); "fears prosecution", "burden, not wanting to abandon the patient" (14); extremely personal choice, I feel that part of some of the resistance at times can be related to [the fact that] that this is something new" "feeling of being torn between professional values and patient values (18); "significant administrative burden" (21); "struggle to reconcile to professional values", sense of responsibility to not create barriers" "walking a tight rope" (22); "tremendous pride", "burden as well" (26); duty to serve. "if not me than who" (27); "interprofessional lack of trust" "excessive workload and lack of financial satisfaction" (29); "burgeoning relationship between palliative care and MAiD", " positive because master of destiny" , "uncomfortable discussing it" (31); "Good palliative care takes a lot of time and interdisciplinary resources. . .when a patient is requesting MAID, most of the resources have been sucked up by that one case … And all of the high-quality palliative care that we do falls by the wayside for the other patients." (35)

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Supplementary appendix 4: PRISMA and ENTREQ checklist.

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	p.4-6
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	p.2 and p.6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	p.6-8
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	p.8-9, supplementary appendix 1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Supplementary appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	p.9
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	p.9-10
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	p.6-7
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	p.6-7
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	p.9-10, supplementary appendix 2
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Not applicable
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	p.10-11
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	p.10-11
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	p.10-11 supplementary appendix 3
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	p.10-11
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Not applicable
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Not applicable

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Supplementary appendix 4: PRISMA and ENTREQ checklist.

Section and Topic	Item #	Checklist item	Location where item is reported
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	high risk, p.20
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	p.5, supplementary appendix 3
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1, p.7
Study characteristics	17	Cite each included study and present its characteristics.	Supplementary appendix 3
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supplementary appendix 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Not applicable
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Supplementary appendix 2
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	p.11-14. Supplementary appendix 3
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Not applicable
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Not applicable
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Not applicable
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 1
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	p.17-20
	23b	Discuss any limitations of the evidence included in the review.	p.17-20
	23c	Discuss any limitations of the review processes used.	p.17-20
	23d	Discuss implications of the results for practice, policy, and future research.	p.17-20
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	p. 11
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	p.11
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	none
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	p.21
Competing interests	26	Declare any competing interests of review authors.	p.21

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Supplementary appendix 4: PRISMA and ENTREQ checklist.

Section and Topic	Item #	Checklist item	Location where item is reported
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	p.21

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71
For more information, visit: http://www.prisma-statement.org/

Enhancing transparency in reporting the synthesis of qualitative research: ENTREQ Checklist (Tong, *et al.,* 2012)

Item No.	Guide and Description	Report Location
1. Aim	State the research question the synthesis addresses	Background, p.6
2. Synthesis methodology	Identify the synthesis methodology or theoretical framework which underpins the synthesis, and describe the rationale for choice of methodology (e.g. meta-ethnography, thematic synthesis, critical interpretive synthesis, grounded theory synthesis, realist synthesis, meta-aggregation, meta-study, framework synthesis)	Data analysis, p.10
3. Approach to searching	Indicate whether the search was pre-planned (comprehensive search strategies to seek all available studies) or iterative (to seek all available concepts until they theoretical saturation is achieved)	search strategy screening and eligibility criteria *SPIDER,* p.6
4. Inclusion criteria	Specify the inclusion/exclusion criteria (e.g. in terms of population, language, year limits, type of publication, study type)	*Eligibility criteria,* p.7
5. Data sources	Describe the information sources used (e.g. electronic databases (MEDLINE, EMBASE, CINAHL, psycINFO), grey literature databases (digital thesis, policy reports), relevant organisational websites, experts, information specialists, generic web searches (Google Scholar) hand searching, reference lists) and when the searches conducted; provide the rationale for using the data sources	search strategy, p.8
6. Electronic Search strategy	Describe the literature search (e.g. provide electronic search strategies with population terms, clinical or health topic terms, experiential or social phenomena related terms, filters for qualitative research, and search limits)	Supplementary appendix 1 and p.6-9
7. Study screening methods	Describe the process of study screening and sifting (e.g. title, abstract and full text review, number of independent reviewers who screened studies)	p.9 study selection process, *Fig 1 PRISMA flow diagram*

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Supplementary appendix 4: PRISMA and ENTREQ checklist.

8. Study characteristics	Present the characteristics of the included studies (e.g. year of publication, country, population, number of participants, data collection, methodology, analysis, research questions)	*Table 2 in supplementary appendix 3, Characteristics of included studies*
9. Study selection results	Identify the number of studies screened and provide reasons for study exclusion (e.g. for comprehensive searching, provide numbers of studies screened and reasons for exclusion indicated in a figure/flowchart; for iterative searching describe reasons for study exclusion and inclusion based on modifications to the research question and/or contribution to theory development)	Fig 1 - PRISMA flow diagram
10. Rationale for appraisal	Describe the rationale and approach used to appraise the included studies or selected findings (e.g. assessment of conduct (validity and robustness), assessment of reporting (transparency), assessment of content and utility of the findings)	Table 1, CERQual approach
11. Appraisal items	State the tools, frameworks and criteria used to appraise the studies or selected findings (e.g. Existing tools: CASP, QARI, COREQ, Mays and Pope [25]; reviewer developed tools; describe the domains assessed: research team, study design, data analysis and interpretations, reporting)	Appraisal of the methodological limitations of included studies, Table 1, CERQual approach
12. Appraisal process	Indicate whether the appraisal was conducted independently by more than one reviewer and if consensus was required	p.10, independently done by the three researchers and consensus achieved.
13. Appraisal results	Present results of the quality assessment and indicate which articles, if any, were weighted/excluded based on the assessment and give the rationale	Table 1, CERQual approach
14. Data extraction	Indicate which sections of the primary studies were analysed and how were the data extracted from the primary studies? (e.g. all text under the headings "results /conclusions" were extracted electronically and entered into a computer software)	Data extraction and analysis, p.10
15. Software	State the computer software used, if any	None used
16. Number of reviewers	Identify who was involved in coding and analysis	
17. Coding	Describe the process for coding of data (e.g. line by line coding to search for concepts)	p.10
18. Study comparison	Describe how were comparisons made within and across studies (e.g. subsequent studies were coded into pre-existing concepts, and new concepts were created when deemed necessary)	Table 2 in supplementary appendix 3.
19. Derivation of themes	Explain whether the process of deriving the themes or constructs was inductive or deductive	Inductive process, p.10

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Supplementary appendix 4: PRISMA and ENTREQ checklist.

20. Quotations	Provide quotations from the primary studies to illustrate themes/constructs, and identify whether the quotations were participant quotations of the author's interpretation	p.12-13
21. Synthesis output	Present rich, compelling and useful results that go beyond a summary of the primary studies (e.g. new interpretation, models of evidence, conceptual models, analytical framework, development of a new theory or construct)	Discussion, p.17-20

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Emotional Impact on Health Care Providers Involved in Medical Assistance in Dying (MAiD): A systematic review and qualitative meta-synthesis.

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Keywords:	MEDICAL ETHICS, MEDICAL LAW, International health services < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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Title: Emotional Impact on Health Care Providers involved in Medical Assistance in Dying (MAiD): A systematic review and qualitative meta-synthesis.

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Key words: Ethics-Medical, Health Personnel, Euthanasia, Right to Die.

Word count: 3996 (excluding title page, abstract, references, figures and tables, acknowledgement, contributory, competing interests, data-sharing and funding statements)

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Abstract:

Title: Emotional Impact on Health Care Providers involved in Medical Assistance in Dying (MAiD): A systematic review and qualitative meta-synthesis.

Background:

Medical Assistance in Dying (MAiD) traverses challenging and emotionally overwhelming territories: Health Care Providers (HCPs) across jurisdictions experience myriad of affective responses secondary to possible tensions between normative and interwoven values, such as sanctity of life, dignity in death and dying, and duty to care.

Objective: To determine the emotional impact on HCPs involved in MAiD.

Methods: Inclusion restricted to English language qualitative research studies from 4 databases (OVID Medline, EMBASE, CINAHL, and Scopus), from beginning until April 30, 2021, and grey literature up to August 2021 were searched. Key author, citation, and reference searches were undertaken. We excluded studies without rigorous qualitative research methodology. Included studies were critically appraised using Joanna Briggs Institute Critical appraisal tool. Analysis was conducted using thematic meta-synthesis. The cumulative evidence was assessed for confidence using the Confidence in the Evidence from reviews of qualitative research (CERQual) approach.

Results: The search identified 4522 papers. Data from 35 studies (393 physicians, 169 nurses, 53 social workers, 22 allied healthcare professionals) employing diverse qualitative research methodologies from 5 countries were coded and analyzed. The thematic meta-synthesis showed three descriptive emotional themes — (1) Polarized emotions including moral distress (n=153), (2) Reflective emotions with MAiD as a 'sense-making process' (n=251), and (3) Professional values-driven emotions (n=352).

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Discussion: This research attempts to answer the question- 'what it means at an emotional level', for a MAiD practitioner. Legislation allowing MAID for terminal illness only influences the emotional impact: MAiD practitioners under this essential criterion experience more polarized emotions, whereas those practicing in jurisdictions with greater emphasis on allaying intolerable suffering experience more reflective emotions. MAiD practitioner's professional values and their degree of engagement influences the emotional impact, which may help structure future support networks. English language literature restriction and absence of subgroup analyses limits the generalizability of results.

Other: Funding source: none.

Key words: Ethics-Medical, Health Personnel, Euthanasia, Right to Die.

Strengths and Limitations of this study:

Strengths:

- An eligibility criteria and subsequent search strategy that focusses on emotional impact of MAiD on HCPs with qualitative research methodology.
- Use of Joanne Brigg's critical appraisal tool for assessment of risk of bias and use of the CERQual approach for assessing the methodological limitations, relevance, coherence and adequacy of the evidence after completion of meta-synthesis.

Limitations:

- Qualitative signals of absence of sub-group analysis, eligibility criteria limited to published English-language literature and fast-moving pace of research on emotional impact of MAiD on HCPs likely contributes to significant publication bias.
- Generalizability of evidence limited by presence of selection bias in included studies.

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Introduction:

Medical Assistance in Dying (MAiD) poses ethically complex challenges that can be a major source of distress to participating Health Care Providers (HCPs) —especially since MAiD may involve navigating conflicting personal and professional values. These values are contextual, dynamic and often not in alignment with each other; for example, professional values of duty to care and reducing suffering in case of terminal illness through MAiD may conflict with the moral value of preserving sanctity of human life, as the later may involve forbidding any action that hastens a patient's death in the dying process (1, 2). In the context of assisted death, a HCP often has to navigate value-conflicts between respect for autonomy and patient right to self-determination vs. respect for individual human life, and human life in general. Except for Switzerland, all other countries require HCPs to be at the forefront in discussing and executing eligible requests for assisted death within their defined jurisdictions (3).

Assisted death in selected jurisdictions-overview and current status

The number of jurisdictions across the world with medically assisted death legislation continues to grow. Switzerland, Netherlands, Belgium, Luxemburg, Canada, besides jurisdictions in the USA (Oregon, Vermont, California, Washington State, Colorado, the district of Columbia, Hawaii, Maine and New Jersey) along side the State of Victoria, Tasmania and South Australia in Australia and Columbia in South America, and most recently Spain and New Zealand, have legalized medically assisted death in some form (3, 4). Assisted death legislations in Canada, the State of Victoria in Australia and the Benelux countries includes both assisted suicide and euthanasia. Jurisdictions in the USA and Switzerland allow only assisted suicide.

Broadly speaking, the 'Benelux' countries (Belgium, Netherlands and Luxemburg) have less restrictive rules in place for MAiD than the American jurisdictions that permit this practice. For example, Benelux countries allow advanced directives, and terminal of illness is not a requirement for MAiD eligibility in Belgium and Netherlands. Jurisdictions in the USA, on the other end, have strict eligibility criteria that the

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illness must be terminal and there must be some timeline to foreseeability of natural death—commonly 6 months in most jurisdictions.

Intact decision-making capacity translating to ability to give informed consent for MAiD, voluntariness of request and suffering from a terminal illness are the mainstay of the eligibility criteria for MAiD, with each criterion receiving variable emphasis, depending on the legislative jurisdiction. For example, "reasonable foreseeability of natural death" criterion was removed from Canada's MAiD eligibility criteria following recent changes in the legislation (5-8).

HCPs and MAiD—current knowledge and knowledge gaps.

From an ethics perspective, amongst the HCPs, the physician's role in providing MAiD is perhaps the most ambiguous. Historically, medicine as a profession is rooted in the ethical principle of 'first, do no harm' while providing care. While this is true, medical futility and the sense of powerlessness and loss of control at end-of-life are a reality in modern medical practice, which is often reflected as physician ambivalence to participate in MAiD (9-11).

While this sense of moral ambiguity may distance physicians from the practice of MAiD, nurses also share the complex attitudes and polarized feelings towards MAiD (12). This complexity is often due to the dual role that nurses play in most health care systems around the world: on one end, they act as a strong advocate for patient's wishes, whereas on the other end, they only have a supportive role in medical decision-making process. A recent synthesis of qualitative studies describing registered nurses' experiences with MAiD from Belgium, the Netherlands, and Canada showed that while the nurses played a central role in providing important 'wrap-around' care for patients and family, their participation in MAiD required significant moral work (13).

A recent scoping review exploring the challenges faced by HCPs while handling MAiD requests found lack of clear guidelines/protocols, role ambiguity, difficulties in evaluating capacity/consent, conscientious

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objection, lack of inter-professional collaboration and difficulties in assessing nature and severity of suffering as major barriers in developing comprehensive care models for implementation of MAiD (14). Furthermore, the scoping review also pointed out that HCPs need substantial degree of time and emotional commitment to participate in a MAiD request. A scoping review and thematic meta-synthesis of qualitative studies exploring HCPs' attitudes towards assisted death practices in Belgium, Netherlands, Israel, Australia, Germany and the USA showed that their attitudes were shaped by a deep sense of moral responsibility and contextual care-relationships (15).

This empirical evidence provides valuable insights on experiences and attitudes of HCPs towards MAiD; however, the nature and extent of emotional impact remains unexplored. Preliminary evidence suggests that the shift away from the medicine's fundamental values of cure/control of disease to hastening death with the aim of alleviating intolerable suffering can have significant emotional effects on participating HCPs. These can range from feeling overwhelmed with a sense of powerlessness on one end, to a rewarding and a positive experience on the other (16, 17).

Objectives: To determine the emotional impact on HCPs involved in MAiD.

Methods:

Search strategy, screening and eligibility criteria:

The inclusion and exclusion criteria were developed in line with SPIDER (Sample, Phenomenon of Interest, Design, Evaluation, Research type) (18). In order to ensure qualitative richness of themes, we included all qualitative research studies and excluded surveys, personal anecdotes, attitudes and experiences without in-depth qualitative analysis published on this topic.

Relevant definitions: For the sake of this review, we define a Health Care Provider as a person "lawfully entitled under the law of a province to provide health services in the place in which the services are

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provided by that person"(19). This definition includes pharmacists, nurses, nurse practitioners, social workers, spiritual health practitioners, psychotherapists and clinical psychologists who are legally authorized to practice within their respective scope of practice. We included 'Assisted suicide assistant' and provider in 'Right to die' societies in Switzerland as unique MAiD care-providers who contact the eligible participant and liaise with the physician and pharmacist in the conduct of MAiD.

For the sake of this review, the term 'MAiD' refers to (20):

a. The administering by a physician or nurse practitioner of a substance to a person, at their request, that causes their death (euthanasia); and/ or

b. The prescribing or providing by a physician or nurse practitioner of a substance to a person at their request, so that they may self-administer the substance and in doing so cause, their own death (assisted suicide).

Eligibility criteria:

1) Includes worldwide published literature on the research question in English language, inclusive of all age groups; articles published up to April 30, 2021.

2) Includes all qualitative studies evaluating the emotional impact through qualitative research methodologies like grounded theory, semi-structure interviews, narrative inquiry or others, and describes/mentions:

    a. 'HCPs' and 'MAiD' as defined above

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b. The emotional impact on HCPs in terms of emotions /affective responses experienced or expressed while accessing, discussing, participating or caring for the patient who has made a valid MAiD request.

3) Excludes case studies, anecdotes or studies without a description or mention of a rigorous qualitative research methodology.

Search strategy:

An iteratively developed search strategy was developed and piloted with the help of 3 librarians with expertise in systematic review search strategies. Considering the inter-disciplinary nature of the objective, the search strategy was conducted on OVID Medline), CINAHL, EMBASE and SCOPUS databases. The search terms included three main domains—MAiD, HCPs and qualitative research methodology and their synonyms. Full search strategy on the 4 databases is available in supplementary appendix 1.

In addition to database searches, the study team conducted a grey literature search (21) which was informed by search methods outlined by Godin et al (22), using the same search terms and their synonyms. Grey literature was retrieved between December 10, 2018, and March 1, 2019, and updated on August 10, 2020, and August 10, 2021, from:

(1) Databases including Google scholar, the Canadian electronic library and the Canadian Institute for Health Information and

(2) OpenGrey, BASE (Bielefeld Academic Search Engine) and the OAIster catalogue of open access resources that includes digital thesis sources like the WorldCat.

The grey literature search strategy and results are included in supplementary appendix 1. For the purpose of feasibility, reports from the year 2000 and beyond were retrieved. In addition, backward citation tracking was conducted by hand searching the reference lists of all included papers.

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Study selection process:

All identified records were imported into the reference management software, Zotero and duplicates removed by the lead researcher (SD). 20% of the title and abstracts of peer reviewed records were independently screened by two reviewers (AS and AB) based on the eligibility criteria; SD screened the remaining 80% for eligibility and reviewed the results with AS and AB in regular team meetings. Given that a substantial portion of grey literature did not include abstracts, the grey literature screening process was initiated at the full-text phase. SD consulted the keywords of yielded academic records if the title and abstract lacked clarity in relation to core concepts and reviewers AB and AS independently assessed any records for any discrepancy and/or uncertainty regarding their inclusion. The researchers met at the beginning, middle and end of the screening process to ensure consistency. SD, AS and AB independently screened the full texts of the academic and grey literature, applying the same inclusion and exclusion criteria in successive team meetings to resolve any discrepancies.

Patient and Public involvement: No patients involved.

Assessment of risk of bias:

We used the Joanna Briggs Institute Critical appraisal tool for use in systematic reviews: checklist for qualitative research to critically appraise the included studies over 10 constructs. These constructs range from congruency to philosophical construct to theoretical and cultural location of the researcher (23). The results of the assessment of risk of bias were independently reviewed by AB and AS and are presented in detail in supplementary appendix 2.

The search results and reasons for exclusion at each stage of screening were recorded and represented in the adapted Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) flow diagram in Figure 1.

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Data Analysis:

Data Extraction and Data analysis:

We adopted a thematic synthesis approach to analyze and synthesize data. Thematic synthesis is an adaptation of thematic analysis and provides a set of established methods and techniques that help synthesize qualitative research outcomes, especially when there is heterogeneity in the outcome variables (24). This approach is especially useful in our case since it enables us to examine the meaning, significance and social constructions around the emotional experience of a HCP involved in MAiD. SD independently coded each line of text according to its meaning and content. Codes were listed as 'free' codes, without any hierarchical structure. AB and AS cross-checked the coded data for any discrepancy. Subsequent thematic synthesis was done by SD, AB and AS in the following 2 stages:

Stage 1: Identifying the similarities between the codes.

All relevant qualitative data from the selected studies were extracted manually from the results, discussion and conclusion section and are represented in Table 2 of supplementary appendix 3. The codes were inductively grouped into descriptive themes so that patterns could be identified.. The use of line-by-line coding enabled us to undertake translation of concepts from one study to another. Based on the similarities and differences of emerging codes, descriptive themes were generated, and each theme was entered as boxes and codes from each study illustrated in those boxes, so that constant comparison analysis process could be done (see Table 3 in supplementary appendix 3).

Stage 2: Development of analytic themes.

In this last stage, the descriptive themes were further interpreted using reciprocal translation and constant comparison methods to develop analytic themes. At this stage, the meaning of the patterns of

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the descriptive themes was analyzed against the research question so that a narrative component could be developed.

Once thematic synthesis was completed, each researcher independently evaluated the cumulative evidence from individual studies for methodological limitations, relevance, coherence and adequacy using the Confidence in the Evidence from reviews of qualitative research (CERQual) approach (see table 1) (25).

All researchers met during regular research-review meetings to resolve any discrepancies and achieve consensus over the assessment.

This systematic review was a part of an academic capstone project and was not registered with any international database. The review protocol is available from the research team on request.

In addition to employing the PRISMA Checklist for systematic reviews, we used the Enhancing transparency in reporting the synthesis of qualitative research (ENTREQ) checklist to improve the reporting of our meta-synthesis (see supplementary appendix 4).

Results:

Characteristics of included studies:

35 qualitative research studies were included in the review. The included literature was based in 5 countries: The United States of America [7], The Netherlands [9], Canada [14], Belgium [1], Switzerland [3], and one study was an international study with participants from the United States of America and Netherlands. The data included 393 physicians, 169 nurses, 53 social workers in hospice care, 11 allied health care professionals (7 personal support workers, 1 pharmacist, 1 genetic technologist and 2 psychologists) and 8 directors of socio-medical institutions and 3 socio-cultural animators (applied sociologists who work along side communities at grass roots to develop and facilitate programs that

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support action for local and social change). A detailed description of the included studies is included in Table 2 of supplementary appendix 3.

Thematic synthesis*:*

Stage 1: Descriptive themes:

Three descriptive emotional themes were derived from the thematic synthesis:

Dimension 1: Strong, internalized and polarized emotions (studies referenced 26-36): These included three subordinate categories/genres of:

- Positive emotions of 'reward', 'relief', 'active openness', 'overwhelming but uplifting' feelings;
- Negative emotions of 'powerlessness', 'guilt', 'emotional exhaustion', 'vicarious suffering' and fear of a slippery slope and losing control and

Individual conscience-based emotions of 'moral shudder' and moral distress. This emotional dimension was strongly embedded in the cultural and political milieu and the interpersonal communication strategies used by the HCP.

Dimension 2: Reflective, discourse-based emotions (studies referenced 26,30,36,37-45): These included emotions of 'growing with the patient's experience', MAiD as a 'sense-making process', 'de-tabooing the philosophical meaning of death through MAiD' and various degrees of 'dynamic conflict' secondary to a reflective sense of insecurity. These emotions were descriptively laid on a platform of 'interpretative therapeutic engagement', where they seemed to aid in the larger philosophical and societal discourse around MAiD (46).

Dimension 3: Emotions that resonate with professional values (studies referenced 28,30,34,39,47-61): These included emotions that resonated with professional values like 'competency and perfection',

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'intimate care', '*colloque singuliar*' (singular language of trust and conscience in context of therapeutic relationship) and various degrees of commitment ranging from 'contractual' to 'sacrificial'.

Table 1 illustrates some of the quotes demonstrating of the descriptive emotional themes.

Table 1: Descriptive themes and illustrative quotes:

Descriptive Theme		Illustrative quotes	Country/Reference
Strong, internalized, and polarized emotion theme	Positive emotions.	I think when you see the patients that we see, it's very clear that you're doing an incredible service. And that's wonderful. There isn't a single moment when I see these patients that I don't think, "Oh my God, I'm so happy to be here to help you." So that's tremendously reinforcing ..."	Canada/Shaw et al., 2018, p.e397.
	Negative emotions	"It was terribly creepy, I never went anywhere with as much lead in my shoes as that morning when I took my bag with the medication in it (T, male)."	Netherlands/ van Marwijk H et al. 2007, p611.
	Moral distress	"There is just a standard that I have. I could not live with myself if I knew that I broke one of the Ten Commandments. I don't feel that I have the right to do that. I will say that there have been times when I would have liked to do that… And there have been times when I've thought about it, and maybe I got right up to the edge. But I wouldn't – I couldn't go over the line…"	USA/Judith Schwarz, 2004, p.229
Reflective emotion's theme		"I shy away from saying suicide or euthanasia. The act of it, however we name it, calls for the most profound respect as the consequence is that a heart stop beating, lungs stop breathing, forever. I am working and sense making as I go along, being sure that I keep breathing."	Canada/ Beuthin R., 2018, p1684
Professional values-driven emotional theme		"Patients have the right to make as many decisions as they are able to make for themselves, and we respect those even though they may not be the same decisions that we might make . . . and we will	

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	advocate for the patients . . . to get them whatever they want . . . I believe in self-determination, but I think it's (PAS) a sad commentary on our society." (Social Worker)	

Stage 2: Analytic themes

Analytic themes in thematic synthesis typically 'go beyond' the findings of the primary studies and generate additional concepts, understandings or hypothesis. At this stage, we used the descriptive themes to answer the review question as to how and why did the HCPs participating in MAiD experience such complex emotions. Each reviewer, initially independently and then as a group, inferred the factors that likely influence the experience of the descriptive themes by questioning how HCPs participating in MAiD represent themselves, or their emotions in the context of their larger health care environment. This process was repeated until the new themes were sufficiently abstract to explain all our initial descriptive themes. Altogether, this process resulted in generation of 2 analytical themes:

1. Legislative emphasis on terminal illness as a necessary inclusion criterion for MAiD influences the emotional impact: In jurisdictions that legislate MAiD with the central aim to alleviate intolerable suffering in context of terminally ill medical conditions (example the USA), the HCPs experience strong polarized emotions that are modulated by their individual cultural/religious background. The extent of emotional impact ranges from positive emotions of reward/relief on one end, to negative (burden, emotional exhaustion) and conscientious based moral distress on the other. This is in sharp contrast to the emotional impact on HCPs in jurisdictions that legislate MAiD with an emphasis on alleviating intolerable suffering without terminal illness being a necessary requirement (for example Benelux countries, Switzerland, and more recently, Canada). The HCPs in these jurisdictions experience the emotional impact of MAiD as a 'sense-making' process—this

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Table 2: Grading of Recommendations, assessment, development and evaluation (GRADE) Confidence in the evidence of reviews of Qualitative Research (CERQual) evidence profile

allows them to reflect on the emotional dissonance between basic emotions and emotions that conform to legislative rules.

2. Values associated with the HCPs' profession and their degree of engagement in the MAiD process are strong influential factors that shape the emotional impact of MAiD. For example, because of their everyday involvement with patients and emphasis on professional values of helping others, compassion and patient advocacy, the emotional impact on nurses involved in MAiD (studies referenced 28, 30, 34-36, 39, 41, 42, 45, 53, 55, 57, 60, 61) demonstrated strong, polarized positive as well as negative emotions. As one nursing participant noted, "…it's the hardest nursing. I've worked [in the emergency department], I've worked medicine floor, this is the hardest nursing there is, having somebody pass away, you actually feel something pulled out of you when that person passes. There's something missing. … If you take care of somebody for an extended time and they pass away, you just feel, I just feel coldness, or whatever. You just feel drained …. (*36, p57*).

Appraising the quality of evidence-the GRADE CERQual approach:

Evidence from qualitative evidence syntheses is increasingly incorporated into decision-making processes and the GRADE CERQual approach allows the user to make a transparent assessment of how much confidence decision-makers and other users can place in individual review findings from syntheses of qualitative evidence. In order to ascertain the degree of confidence, we graded the evidence in terms of adequacy, relevance, coherence as well as methodological limitations using the GRADE CERQual approach (25). Table 2 illustrates a summary of the findings and the GRADE CERQual profile.

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Summary finding	Studies contributing substantially to the summary theme (studies numbered as per Table 2 in supplementary appendix 3)	Methodological Limitations	Coherence	Adequacy	Relevance	CERQual GRADE evidence	Explanation of CERQual assessment
HCPs experienced strong, internalized, often polarized and deeply personal basic emotions that were modulated by the HCP's cultural and/or religious background. **Level embedded**: cultural/religious	1,2,3,5,13,18, 19,21, 24,26,28.	Minor methodological limitations concerning location of the researcher theoretically/ culturally, And influence of the researcher on the research and vice versa	Moderate concerns regarding coherence	Minor concerns regarding adequacy	No or very minor concerns regarding relevance	High	Variability in experiences of participants posed a challenge with respect to coherence, however, this also added to the richness of results. Hence, we have graded the confidence in quality of findings as high.
Influenced by the socio-political environment as well as the social discourse on suffering and death, HCPs shared emotions of personal growth/sense-making and relational experiences of	2,5,6,8,11,14, 23,25, 28,30, 32, 34.	Moderate/min or methodological limitations concerning location of the researcher theoretically/ culturally, and influence of the researcher on the research	No or very minor concerns regarding coherence	No or very minor concerns regarding adequacy	No or very minor concerns regarding relevance.	High	Paper 6 did not approach the ethics committee and hence does not have ethics committee approval. Apart from this study, all studies in this group

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deeper compassion and sympathy. HCPs also experienced emotional dissonance over personal emotions and emotions expressed to conform to legislative rules. **Level embedded**: Socio-political		and vice versa					contributed to the summary findings in terms of coherence, adequacy and relevance. Hence, we have graded the confidence in the quality of the findings to be high.
HCPs expressed emotions aligned with their individual professional values and beliefs systems and, most of the times, attempted to align their values associated with the MAiD ideology; at other times, legislation of respective jurisdictions helped shape emotional experiences. **Level embedded**: Professional/ legal	3,4,5,7,9,10,12, 15,16, 17,20,22,24,27, 29,31, 33,35.	Moderate Methodological limitations concerning location of the researcher theoretically/culturally, And influence of the researcher on the research and vice versa. Also, selection of participants Paper 16, one single hospital.	minor concerns regarding coherence	No or very minor concerns regarding adequacy	No or very minor concerns regarding relevance	Moderate	Most of the studies in this group had methodological problems of selection bias and lack of generalizability. For example, paper 16 selected participants from a single hospital-based setting. The findings are limited in terms of generalizability to similar groups in different settings. Hence, we have graded

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							down our confidence in the quality of findings to moderate.

Discussion:

Difference in MAiD legislation in Benelux and Non-Benelux countries-key features:

The substantive and procedural requirements for MAiD across global jurisdictions rests on 3 main pillars: patients' right for self-determination expressed through voluntariness of request and a valid, informed consent process, foreseeableness of natural death due to terminal medical illness and subjective nature of individual suffering (62,63). The key difference between the legislations for MAiD in Benelux countries and countries like the USA is the differential emphasis on eminent or foreseeableness of death. The MAiD legislations in Belgium, Netherlands, Switzerland, and, more recently Canada have a more permissive legal framework that allows people to access MAiD as a service to end their intolerable suffering that has no prospect of improvement but is not necessarily terminal.

MAiD Legislation and its shaping effect on the emotions of the involved HCP:

An important take home message from this evidence synthesis is how legislations have a shaping effect on emotional responses. The HCPs who practice in the Benelux countries and Switzerland seem to experience more reflective emotions over strong polarizing emotions expressed by HCPs who practice in non-Benelux countries like the USA. Canada seems to have a unique, transitional position—with the emphasis of the legislation going the Benelux countries' way, the HCPs emotional experiences show a mixture of emotions driven by their professional values as well as the ongoing societal discourse on MAiD. This observation conforms to Michel Foucault's position on how law acts as an element in the expansion

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of power(s) (64); legislatures along with other platforms of knowledge expression modulate every fiber of human society. Our thematic synthesis points out that the Law that limits application of MAiD to terminally illnesses provide for a broader range of emotional expression. Thus, legislation on MAiD across the globe provides the HCP with a locus of administrative control which then decides how the emotional discourse around MAiD is shaped; the question is—*how* do we want the *emotional discourse* around MAiD to be shaped?

MAiD legislation, societal values, and emotional impact on the involved HCP: A complex relationship.

On one end, attitudes of physicians towards MAiD has shown reflective trends to legislative standards; countries like Belgium and Netherlands find much stronger physician support than their USA counter parts (65). On the other end, public support towards MAiD has been reflective of the prevailing societal cultural and religious practices; central and eastern European countries have shown a decline in support with corresponding increase in religiosity as opposed to western European countries (66,67). While an assisted-death legislation with its rules and safeguards provides an obligatory, 'top-down' framework to embed MAiD within health care, it does not necessary reflect the integration of MAiD within the value-based relationships that have traditionally defined an individual's health care (68). Hence, although a MAiD legislation to integrate MAiD into health care is a likely reflection of a consensus position of a society, it does challenge the moral environments within which HCPs practice medicine, thereby influencing the emotional impact on HCP. HCPs subsequent attempt to align themselves with their own professional values, legislative standards and public perceptions can lead to intense emotional responses, both, within their internal, personal and their external professional spaces.

Emotional discourse amongst HCPs involved in MAiD: HCP role and ethics of Care

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The right to choose when and how to die has always been a contentious issue across various societies (69-71). Public discourse on MAiD are shaped through societal emphasis on individual as well as contextual factors associated with assisted death—these often range from religious beliefs regarding sanctity of human life and personal meaning of death to loss of autonomy associated with illness-related intolerable suffering. With advancing medical technologies, the potential to prolong life has increased significantly (72,73), and the HCPs assumes a central position to shape the discourse around assisted death.

In countries where MAiD is legalized but is restricted to terminal illnesses with imminent chance of death, the position of a HCP continues to be one that of a provider of 'Care'. Here, the moral dimension of 'Care' continue to be defined as 'everything we do to maintain, continue or repair our world so that we can live in it as well as possible' (74). The value of care in health care systems have been traditionally associated with attentiveness, responsibility, nurturance, compassion and meeting others' needs (75). While emotional responses to legal requests of hastening death is affected by policies, professional identity, commitment to patient autonomy, personal values and beliefs, the patient-clinician relationship and will vary on a case-by-case basis (76), this systematic review raises an important question—How does legalizing MAiD with emphasis on alleviating intolerable suffering without the context of a terminal illness change the moral dimensions of Care?

Conclusion:

HCPs involved in MAiD experience a myriad of emotions that includes positive/negative emotions, reflective, 'sense-making' emotions and/or professional value driven emotions. Emphasis on terminal illness only as an essential criterion, MAiD practitioner's individual professional values and their degree of engagement influence this rich and diverse emotional discourse.

Limitations of the review:

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This review is limited by its focus of emotional impact on HCPs only and the obvious selection bias in the included studies—those who could and volunteered to express their emotions are represented in the review. The review is also limited with absence of sub-group analysis with respect to HCPs' age, years of experience and the influence of gender on the results. Restriction to English language studies likely carries a high risk of publication bias.

There are several gaps in our understanding of the emotional impact on HCPs involved in MAiD that would benefit from further research. Intolerable suffering is a common eligibility requirement for assisted death, although HCPs often struggle to understand and assess the nature and normative function of suffering. Is it the very nature of the emotional tone of suffering which is overwhelming or is it more to do with what lies underneath that makes suffering 'intolerable'? Is there room for humanistic narratives around meaning behind and endurance of one's suffering? Such questions confront MAiD practitioners and an in-depth exploration of this nebulous concept of intolerable suffering in context of assisted death may help HCPs navigate their emotional experience while providing MAiD.

Ethics statement: This is a systematic review and meta-synthesis of already published and accessible research data and does not require ethics committee or Institutional board approval.

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Contributorship Statement: The authors confirm contribution to this systematic review and meta-synthesis as follows:

- study conception and design: Dr. Saumil Dholakia, Dr. Alireza Bagheri, Dr. Alexander Simpson.

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- development of eligibility criteria: Dr. Saumil Dholakia, Dr. Alireza Bagheri and Dr. Alexander Simpson.
- search strategy developed by Erica Lenton, Heather Cunningham (Library services, Gerstein's library, University of Toronto) and Fiona Inglis (Library services, Center for Addiction and Mental Health, University of Toronto) in close consultation with Dr. Saumil Dholakia and reviewed by Dr. Alireza Bagheri and Dr. Alexander Simpson.
- Study selection and data extraction process by Dr. Saumil Dholakia and independently reviewed by Dr. Alireza Bagheri and Dr. Alexander Simpson.
- Dr. Saumil Dholakia performed the assessment of risk of bias, which was independently reviewed by Dr. Alireza Bagheri and Dr. Alexander Simpson.
- All three authors were involved equally in performing the qualitative meta-synthesis and CERQual assessment.
- Draft manuscript preparation: Dr. Saumil Dholakia with multiple reviews, feedback and edits in form as well as content by Dr. Alireza Bagheri and Dr. Alexander Simpson.
- All authors reviewed the results and approved the final version of the manuscript.

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No unpublished data.

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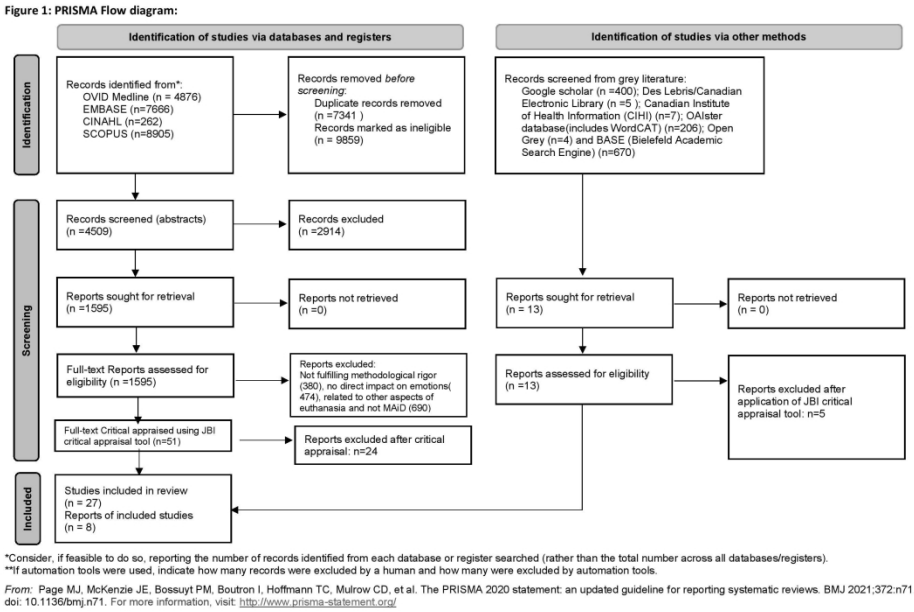
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Figure 1: PRISMA flow diagram: The PRIMSA diagram details our search and selection process applied during the review.



Caption : Figure 1: PRISMA flow diagram: The PRISMA diagram details our search and selection process applied during the review.

296x210mm (250 x 250 DPI)

Supplementary appendix 1:

Database: Ovid MEDLINE(R) ALL <1946 to October 04, 2021>
Search Strategy:

1 euthanasia, active/ or euthanasia, active, voluntary/ or hospice care/ or suicide, assisted/ (14808)
2 terminally ill/ (6684)
3 Right to die/ (4950)
4 Terminal care/ (29907)
5 advance care planning/ or advance directives/ (9125)
6 ((dying or death or euthan* or suicide or terminal* ill*) adj5 (assist* or hasten*)).tw,kf. (5952)
7 Palliative care/ (58012)
8 exp Practice Patterns, Physicians'/es [Ethics] (812)
9 physician's role/ (30584)
10 Health Personnel/ (52294)
11 ((health care provider or clinician* or doctor* or physician* or nurse or social work* or oncologist* or palliative physician or nursing or psychiatrist* or psychologist* or psychotherapist*) adj3 (experience* or emotion* or feeling*)).tw,kf. (23976)
12 (Interview: or experience:).mp. or qualitative.tw. (1655368)
13 health personnel/ or allied health personnel/ or anesthetists/ or caregivers/ or case managers/ or "coroners and medical examiners"/ or emergency medical dispatcher/ or epidemiologists/ or faculty, medical/ or faculty, nursing/ or health educators/ or health facility administrators/ or medical chaperones/ or medical laboratory personnel/ or medical staff/ or exp nurses/ or nursing staff/ or occupational therapists/ or personnel, hospital/ or pharmacists/ or physical therapists/ or physician executives/ or exp physicians/ (431546)
14 (ethnograph* or grounded theory or qualitative research or thematic analysis or semi-structured interview* or narrative inquiry or focus* group or content analysis or discourse or lived life experience*).tw,kf. (156494)
15 aid in dying.mp. (243)
16 death with dignity.mp. (607)
17 Bill C-14.mp. (24)
18 Bill C-7.mp. (2)
19 MAID.mp. (458)
20 physician assisted death.mp. (309)
21 physician assisted dying.mp. (142)
22 (assisted suicide or physician assisted suicide).tw,kf. (3163)
23 Qualitative Research/ (67825)
24 1 or 2 or 3 or 4 or 5 or 6 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 (55486)
25 7 or 8 or 9 or 10 or 11 or 13 (527655)
26 12 or 14 or 23 (1692068)
27 24 and 25 and 26 (5490)
28 limit 27 to dt=19460101-20210430 [January 1st, 1946 to April 30th, 2021] (5441)
29 limit 28 to english language (5073)
30 limit 29 to abstracts (4876)

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Grey Literature databases (December 10th 2018 to March 1st, 2019, updated August 2020 and 2021)

Total # Records after de-duplication: 13. Records selected after applying critical appraisal tool 8.

Database	Search strategy	#records screened	# new records and records after de-duplication and applying the critical appraisal tool
Google scholar	With the exact phrase: "Medical assistance in dying" ; "physician assisted suicide"; With all the words: "emotional impact on health care providers involved in medical assistance in dying"	400	5
Des Lebris/Canadian Electronic Library	Medical assistance in dying	5	0
Canadian Institute of Health Information (CIHI)	Medical assistance in dying	7	0
OAIster database (includes WordCAT)	Medical Assistance in dying, Physician assisted suicide as key word	206	2
OpenGrey	Medical assistance in dying, Physician Assisted suicide as key word	4	0
BASE (Bielefeld Academic Search Engine)	Subject Heading search: "Medical Assistance in dying"	670	1

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Selected records:

Google scholar included Results:

1. Beuthin R. Cultivating Compassion: The Practice Experience of a Medical Assistance in DyingCoordinator in Canada. *Qualitative Health Research*. 2018;28(11):1679-1691. doi:10.1177/1049732318788850
2. Bouthillier M-E, Opatrny L. A qualitative study of physicians' conscientious objections to medical aid in dying. *Palliative Medicine*. 2019;33(9):1212-1220. doi:10.1177/0269216319861921
3. Bruce A, Beuthin R. Medically Assisted Dying in Canada: "Beautiful Death" Is TransformingNurses' Experiences of Suffering. *Canadian Journal of Nursing Research*. June 2019. doi:10.1177/0844562119856234
4. Alison A. Townsley. Putting a Bow on Death and Dying-Health Care Professionals' Experiences with Medical Assistance in Dying (MAiD): A Foucauldian Discourse Analysis with Agambian Insights. [Practice Based Research Paper on the Internet]. York University School of Social Work;2018. [Cited February 28,2019] Available from: https://yorkspace.library.yorku.ca/xmlui/bitstream/handle/10315/35613/Alison_Townsley_PRP_2018.pdf?sequence=1
5. Gamondi C, Borasio GD, Oliver P, Preston N, Payne S. Responses to assisted suicide requests: aninterview study with Swiss palliative care physicians. BMJ Support Palliat Care. 2019 Mar;9(1):e7. doi: 10.1136/bmjspcare-2016-001291. Epub 2017 Aug 11. Available from https://pubmed.ncbi.nlm.nih.gov/28801317/

OAIster included Results:

1. Implementing Medical Aid-in-Dying in Vermont: a Qualitative Study. *J GEN INTERNMED* **34,** 636–641 (2019). https://doi.org/10.1007/s11606-018-4811-1
2. Allyson Oliphant. "If Not Me, Then Who?" The Narratives of Medical Aid in Dying (MAiD) Providers and Supporters Around Their Professional Identity and Role in MAiD [Thesis on theInternet]. MacMaster University; 2017. [Cited on February 10, 2019]. Available from http://hdl.handle.net/11375/22146

BASE included results:

1. Sheridan, Laura, "When Patients Ask to Die: The Role of Nurses in Medical Assistance in Dying"(2017). Electronic Thesis and Dissertation Repository. 5041. https://ir.lib.uwo.ca/etd/5041

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Database:

Embase <1974 to 2021 April 30>

#	Query	Results from search strategy run on October 4, 2021
1	euthanasia, active/ or euthanasia, active, voluntary/ or hospice care/ or suicide, assisted/	18,815
2	terminally ill/	8,339
3	right to die/	4,060
4	terminal care/	38,968
5	advance care planning/ or advance directives/	13,209
6	((dying or death or euthan* or suicide or terminal* ill*) adj5 (assist* or hasten*)).tw,kf.	7,430
7	palliative care/	83,687
8	exp clinical practice/ and medical ethics/	5,575
9	physician's role/	49,149
10	health personnel/	168,037
11	((health care provider or clinician* or doctor* or physician* or nurse or social work* or oncologist* or palliative physician or nursing or psychiatrist* or psychologist* or psychotherapist*) adj3 (experience* or emotion* or feeling*)).tw,kf.	34,589
12	(interview: or experience:).mp. or qualitative.tw.	2,361,122
13	health personnel/ or allied health personnel/ or anesthetists/ or caregivers/ or case managers/ or "coroners and medical examiners"/ or emergency medical dispatcher/ or epidemiologists/ or faculty, medical/ or faculty, nursing/ or health educators/ or health facility administrators/ or medical occupational therapists/ or personnel, hospital/ or pharmacists/ or physical therapists/ or physician executives/ or exp physicians/	1,402,853
14	(ethnograph* or grounded theory or qualitative research or thematic analysis or semi-structured interview* or narrative inquiry or focus* group or content analysis or discourse or lived life experience*).tw,kf.	203,129
15	aid in dying.mp.	293
16	death with dignity.mp.	655
17	Bill C-14.mp.	32
18	Bill C-7.mp.	4
19	MAID.mp.	667

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20	physician assisted death.mp.	365
21	physician assisted dying.mp.	171
22	(assisted suicide or physician assisted suicide).tw,kf.	3,620
23	qualitative research/	98,864
24	1 or 2 or 3 or 4 or 5 or 6 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22	73,993
25	7 or 8 or 9 or 10 or 11 or 13	1,525,230
26	12 or 14 or 23	2,406,823
27	24 and 25 and 26	8,659
28	limit 27 to (abstracts and english language and yr="1946 - 2021")	7,666

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MY

EBSCOhost

CINAHL search strategy: Monday, October 4, 2021 3:59:32 PM

#	Query	Limiters/Expanders	Last Run Via	Results
S25	S22 AND S23 AND S24	Limiters - Published Date: 19460401-20210430; Exclude MEDLINE records; Publication Type: Abstract; Language: English Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S24	S10 OR S12 OR S21	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S23	S6 OR S7 OR S8 OR S9 OR S11	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display

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S22 (S1 OR S2 OR S3 OR S4 OR S5 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20)	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S21 qualitative research	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S20 TX (assisted suicide or physician assisted suicide)	Expanders - Apply related words; Apply equivalent subjects	Interface - EBSCOhost Research Databases Search Screen - Advanced	Display

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		Search		
Search modes - Boolean/Phrase				
		Database - CINAHL		
S19	physician assisted dying	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S18	physician assisted death	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S17	MAID	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S16	Bill C-7	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S15	Bill C-14	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S14	death with dignity	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S13	aid in dying	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S12	TX (ethnograph* or grounded theory or qualitative research or	Expanders - Apply related words; Apply equivalent subjects	Interface - EBSCOhost Research Databases Search Screen - Advanced	Display

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thematic synthesis or semi-structured interview* or narrative inquiry or focus* group or content analysis or discourse or lived life experience*)	Search modes - Boolean/Phrase	Search Database - CINAHL	
S11 health personnel or allied health personnel or anesthetists or caregivers or case managers or "coroners and medical examiners" or emergency medical dispatcher or epidemiologists or faculty, medical or faculty, nursing or health educators or health facility administrators or medical chaperones or medical laboratory personnel or medical staff or nurses or nursing staff or occupational therapists or personnel, hospital or pharmacists or physical therapists or physician executives or physicians	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S10 TX (interview: or experience:) or qualitative	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S9 TX ((health care provider or clinician* or doctor* or physician* or nurse or social work* or oncologist* or palliative physician or nursing or psychiatrist* or psychologist* or psychotherapist*) N3 (experience* or emotion* or feeling*))	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display

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S8 health personnel or healthcare professionals or healthcare workers	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S7 physician role	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S6 practice patterns, physicians AND medical ethics	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S5 TX ((dying or death or euthan* or suicide or terminal* ill*) N5 (assist* or hasten*))	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S4 advance care planning or end of life planning or advance directive or advance care plan or advance decision or advance helth care plan	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S3 terminal care or palliative care or end of life care or hospice	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S2 terminally ill	Expanders - Apply related words; Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	Display
S1 euthanasia or assisted suicide or right to die or physician assisted suicide	or death with dignity	Expanders - Apply related words; Apply equivalent	subjects

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Total Number of records from CINAHL search strategy: 262.

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Scopus Search Sources Lists SD

Saved searches

Combine queries... e.g. #1 AND NOT #3

ID	Name	Query	Documents	Date last run	Actions
#33	scopus 6	(({active euthanasia} / OR {vol untary euthanasia} / OR {hospi View More	8,905	oct 4,2021	
#32	scopus 5	(({active euthanasia} / OR {vol untary euthanasia} / OR {hospi View More	9,662	oct 4, 2021	
#31	scopus 4	(({active euthanasia} / OR {vol untary euthanasia} / OR {hospi View More	9,711	oct 4, 2021	
#29	Scopus	(({active euthanasia} / OR {vol untary euthanasia} / OR {hospi View More	10,097	oct 4, 2021	
#28	3	... View More	852,122	oct 4, 2021	
#27	2	({palliative care} /) OR ({clinic al practice} AND {physician eth View More	1,458,970	oct 4, 2021	

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ID	Name	Query	Documents	Date last run	Actions
#26	1	({active euthanasia} / OR {voluntary euthanasia} / OR {hospice	127,545	oct 4, 2021	
#25	terminally ill terminal care	{terminally ill} OR {terminal care}	72,266	oct 4, 2021	
#24	qualitative research	{qualitative research}	570,500	oct 4, 2021	
#23	physician assisted suicide	{physician assisted suicide}	1,365	oct 4, 2021	
#22	physician assisted dying	{physician assisted dying}	90	oct 4, 2021	
#21	physician assisted death	{physician assisted death}	153	oct 4, 2021	
#20	MAiD	{MAiD}	11,549	oct 4, 2021	
#19	Bill C-7	{Bill C-7}	38	oct 4, 2021	
#18	Bill C-14	{Bill C-14}	120	oct 4, 2021	
#17	death with dignity	{death with dignity} ... View More	2,743	oct 4, 2021	
#16	aid in dying	{aid in dying} ... View More	823	oct 4, 2021	
#15	qualitative methods	TITLE-ABS-KEY (ethnograph* OR "grounded theory" OR "the	439,839	oct 4, 2021	

... View More

ID	Name	Query	Documents	Date last run	Actions
#14	health care provider	TITLE-ABS-KEY ("allied health personnel" OR anesthetists OR	1,203,712	oct 4, 2021	
#13	qualitative interview qualitative study	TITLE-ABS-KEY ("qualitative interview" OR "qualitative study")	119,016	oct 4, 2021	
#12	health care provider experience	(("health care provider" OR clinician* OR doctor* OR physicia	85,449	oct 4, 2021	
#11	health personnel	{health personnel} /	195,852	oct 4, 2021	
#10	physicians role	{physician's role} /	31,290	oct 4, 2021	
#9	clinical practice physician ethics	{clinical practice} AND {physician ethics}	25	oct 4, 2021	
#8	palliative care	{palliative care} /	185,455	oct 4, 2021	
#7	assisted death	((dying OR death OR euthan* OR suicide OR "terminal* ill*"	12,119	oct 4, 2021	
#6	advance care planning	{advance care planning} / OR {advance directives} /	27,184	oct 4, 2021	
#5	right to die	{right to die} /	8,046	oct 4, 2021	
#3	active euthanasia	{active euthanasia} / OR {voluntary euthanasia} / OR {hospice c	38,053	oct 4, 2021	

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Table 2: Critical appraisal of studies using The Joanna Briggs Institute Critical appraisal tool for Qualitative Research:

Study (location, number and category of Participants)	JBI Check list	Q1:Congruity b/w philosophy and research	Q2: Congruity b/w research method and question	Q3:congruity b/w Research method &Data collection	Q4:congruity b/w Research method & analysis	Q5: congruity b/w Research method & Results	Q6: Statement Locating the researcher	Q7: Influence Of Researcher addressed	Q8: Adequate Representa-tion of Participants	Q9:ethical approval	Q10: Conclusion Flows from analysis	Appraisal	R for exclusion
1.Voorhees et al., US and Netherlands, physicians 23		Y	Y	Y	Y	Y	Y	N	Y	Y	Y	Include	
2.Van Marwjik et al. Netherlands 22 Primary care ph		Y	Y	Y	Y	Y	N	N	Y	Unc	Y	Include	
3. Denier Yyonne et al. 2010. Belgium Nurses-n=18		Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
4. Elizabeth Norton et al. 2012 USA-social worker-9		Y	Y	Y	Y	Y	N	N	Y	Unc	Y	include	
5. JJ Georges et al. 2008. Netherlands 30 GPs		Y	Y	Y	Y	Y	N	N	Y	Unc	Y	Include	
6. Snijdewind et al., 2014 (Netherlands, 28 physicians)		Y	Y	Y	Y	Y	N	N	Y	N	Y	include	
7. Katja ten Cate et al., 2017-33 physicians netherlands		Y	Y	Y	Y	Y	N	N	Y	Y	Y	Include	
8. Donald G Van tol et al., 2012. Netherlands 15 physicians		Y	Y	Y	Y	Y	N	N	Y	N	Y	include	
9.Veronica Lorraine Fausto Melchor, 2018. USA Hospice social worker 8		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	include	
10. Pamela Miller et al., 2008 Oregon SW-8		Y	Y	Y	Y	Y	N	N	Y	Unc	Y	Include	
11. Deborah Volker et al., 2001. USA Onco Nurse-40		Y	Y	Y	Y	Y	N	N	Y	Unc	Y	Include	

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12.michael Young et al., 2008 Canada nurses-22	Y	N	Y	Y	Y	N	N	Y	Y	Y	**exclude**	Study done at a time assisted death not legal, so does not meet inclusion criteria.
13. Rosanne Beuthin et al., 2018 Canada nurses-17	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
14.eva Bolt et al., 2017 Netherlands paediatrician-8	Y	Y	Y	Y	Y	N	N	Y	NR	Y	Include	
15.Dolores Angela Castelli Dransart et al., 2017 Switzerland-20 nurse, 1 physician, 8 directors, 3 socio-cultural animators.	Y	Y	Y	Y	Y	N	N	Y	Y	Y	Include	
16. Marianne Dees et al., 2012 Netherlands-phy-28	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
17. Theresa Harvath et al., 2006. USA hospice social workers-20	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
18. Ina Otte et al., 2017. Switzerland GP's-20	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
19. Ada van de Scheur, Arie van der Arend, 1998 Netherlands Nurse-20	Y	Y	Y	Y	Y	N	N	Y	Unc	Y	include	
20.Belanger E.et al., 2019 Canada-palliative care physicians-18	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
21. Jessica Shaw et al., 2018. canada phy-8	Y	Y	N	Y	Y	N	N	Y	Unc	Y	Include	
22. Judith Schwartz 2004. USA nurses-10	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	

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23. Dobscha SJ et al., 2004. USA phy-35	Y	N	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	No theme of emotional impact.
24. Galusko et al., 2015, Germany 19 specialized palliative care physicians.	Y	Y	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	Desire to hasten death-definition ambiguous
25. Susanne Brauer et al., 2015. Switzerland, 12 physicians	Y	Y	Y	Y	Y	N	N	N	N	Unclear	**Exclude**	Opinions known, but no emotional ipact theme
26. Linda (b) Oregon phy-35	N	Y	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	Physician opinion of patients req
27.Deborah-texas nurses-36	N	Y	N	Y	Y	N	N	Y	Unc	Y	**Exclude**	No of the nurses participated in assisted suicide in any way
28. D Van Rooyan, Dutch nurses-7	N	N	Y	Y	Y	N	N	Y	N	Y	**Exclude**	More with withdrawal of treatment does not meet criteria
29. vanderspank canada Nurses	N	N	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	SR on nurses experience with withdrawal of treatment-does not meet criteria
30. Joanne Wolfe USA 324 Oncologists	Y	N	N	Y	Y	N	N	N	Y	Y	**Exclude**	Telephone based survey interviews.
31. Booij et al., 2012 Netherlands 15 physicians	Y	N	Y	Y	Y	N	N	N	Y	Y	**Exclude**	No particular description of emotional impact
32. Denier et al., 2010 Belgium 18 Nurses	Y	N	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	More about communicati

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												on and communicati on attitudes and not about emotional impact
33. Bernadette Dierckx 2010 Belgium 18 nurses	Y	N	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	Stage of carrying out a request, no emotional impact described.
34. sercu et al. 2012	Y	N	Y	Y	Y	N	N	Y	Y	Y	**Exclude**	Palliative sedation and euthanasia-boundry lines unclear in the paper.
35. Volker 2007 USA. 19 oncology advanced practice nurses	Y	N	Y	Y	N	N	N	Y	Y	Y	**Exclude**	No engagement in assisted death as illegal in the place of practice.
36. Thulesius et al. 2013 Sweden	Y	N	Y	Y	N	N	Y	N	Y	Y	**Exclude**	No engagement, assisted death is illegal in Sweden. Majority data from HCPs in Sweden.
37. Marike E. de Boer 2011 Netherlands.	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	Experiences, but no emotional impact
38. Neel De Bal 2006 Belgium	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	Conducted at a time when Euthanasia was still illegal, hence does not meet

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												inclusion criteria.
39. Bernadette 2006 Belgium	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	As above.
40. Veerport et al 2006 USA	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	As above
41. Wright et al., 2017 Canada	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	Data collected in 2012-2013 when MAiD illegal.
42. Curry et al., 2000 USA, Connecticut 909 physicians.	Y	N	Y	Y	N	N	N	N	Y	Y	**Exclude**	Assisted suicide illegal, Plus experiences and no emotional impact
43. Susan Price 2001 USA, 11 nurses and 10 physicians. North Carolina	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	Assisted suicide illegal in North Carolina, hence does not meet inclusion criteria
44. France Norwood 2009 Netherlands	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	No emotional impact. Evaluates absence of abuse
45. Smith et al., 2013 USA, South Mississippi	Y	N	Y	Y	N	N	Y	Y	Y	Y	**Exclude**	Assisted death illegal in mississippi and hence does not meet inclusion criteria
46. Beuthin et al., 2020 Canada 8 physicians.	Y	Y	Y	Y	N	N	Y	Y	Y	Y	include	
47. Khosnood et al., 2018 19 physicians, Canada	Y	Y	Y	Y	N	N	Y	Y	Y	Y	Include	

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48. Pesut et al., 2020 59 RN and NPs, Canada	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	include	
49. Keri-Lyn Durant and Katherine kortes Miller 2020 Canada	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	Include	
50. Snijdewind et al., 2016 Netherlands 28 physicians	Y	Y	Y	Y	N	N	Y	Y	Y	Y	include	
51. Mathews et al., 2021. Canada 23 palliative care providers (13 physicians, 10 nurses)	Y	y	Y	y	N	N	Y	Y	Y	Y	include	

Grey literature	JBI Checklist	Q1:Congruity b/w philosophy and research	Q2:congruity b/w research method &question	Q3:congruity b/w Research method &Data collection	Q4:congruity b/w Research method & analysis	Q5: congruity b/w Research method & Results	Q6:Statement Locating the researcher	Q7: Influence Of Researcher addressed	Q8: Adequate Representa-Tion of Participants	Q9:ethical approval	Q10: Conclusion Flows from analysis	Appraisal	Reason for exclusion
1.Rosanne Beuthin, 2018. Canada. Nurse-1		Y	y	Y	Y	Y	N	N	Y	Y	Y	include	
2. Evenblij et al., 2019. Netherlands, 1374 phys		Y	Y	Y	Y	Y	N	N	N	Y	Y		Exclude. 12 page quest lonnaire survey. Indepth expl Oration of emoti Impact.
3. Alison Townsley 2018, Canada. 7 Nurses, SW and PSWs.		Y	Y	Y	Y	Y	Y	N	Y	Y	Y	include	
4. Gamondi et al., 2017, Switzerland. 23 palliative care physicians		Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
5. Bouthillier M-E, Opatrny L 2019		Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
6. Joy Cayetano-Penman et al., July 2021 Australia. 21 Nurses.		Y	Y	Y	Y	Y	N	N	Y	Y	Y		Exclude. More al Knowledge And attitude Rather than emo Impact.

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7 Bruce A, Beuthin R 2019.	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	
8. Buchbinder et al, 2019 Vermont, USA. 37 health Care providers	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	Include	
9 Allyson Oliphant 2017, Canada	Y	Y	Y	Y	Y	Y	N	Y	Y	Y	include	
10 Kopchek, Lauren 2020, Canada. (masters thesis or 10 Palliative care Nurses)	Y	Y	Y	Y	Y	N	N	N	Y	Y		Exclude as study explored the ethical decis making experiences rather than emotional impact.
11 Gaignard, ME., Hurst, S. 2019. Switzerland. 26 Palliative as well as Primary care Providers.	Y	Y	Y	Y	Y	N	N	Y	Y	Y		Exclude. Explores Perspective on Existential Suffering and Not on Emotional Impact.
12 Ellen Wiebe et Al, April 2021. Canada. 14 Physicians and 1 NP	Y	Y	Y	Y	Y	N	N	Y	Y	Y		Exclude. Themes of Difficulties Providing MAiD during Covid 19 Rather than emo Impact.
13. Sheridan, Laura 2017	Y	Y	Y	Y	Y	N	N	Y	Y	Y	include	

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Total Included studies (N): 27 (Databases: OVID Medline, EMBASE, CINAHL, SCOPUS) + 8 (Grey literature search)=35

Critical Appraisal tool: The Joanna Briggs Institute Critical appraisal tools for use in Systematic reviews: checklist for Qualitative research Available from:
http://joannabriggs.org/research/critical-appraisal-tools.html

Discussion: Lockwood C, Munn Z, Porritt K. Qualitative research synthesis: methodological guidance for systematic reviewers utilizing meta-aggregation. Int J Evid Based Healthc. 2015;13(3):179–187.

Grey Literature databases (December 10th 2018 to March 1st, 2019, updated August 2020 and August 2021)

Total # Records after de-duplication: 13. Records selected after applying critical appraisal tool: 8.

Database	Search strategy	#records screened	# new records selected after applying after de-duplication and applying critical appraisal tool
Google scholar	With the exact phrase: “Medical assistance in dying” ; “physician assisted suicide”; With all the words: “emotional impact on health care providers involved in medical assistance in dying”	400	5
Des Lebris/Canadian Electronic Library	Medical assistance in dying	5	0
Canadian Institute of Health Information (CIHI)	Medical assistance in dying	7	0
OAIster database (includes WordCAT)	Medical Assistance in dying, Physician assisted suicide as key word	206	2
OpenGrey	Medical assistance in dying, Physician Assisted suicide as key word	4	0
BASE (Bielefeld Academic Search Engine)	Subject Heading search: “Medical Assistance in dying”	670	1

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Selected records:

Google scholar included Results:

1. Beuthin R. Cultivating Compassion: The Practice Experience of a Medical Assistance in Dying Coordinator in Canada. *Qualitative Health Research*. 2018;28(11):1679-1691. doi:10.1177/1049732318788850

2. Bouthillier M-E, Opatrny L. A qualitative study of physicians' conscientious objections to medical aid in dying. *Palliative Medicine*. 2019;33(9):1212-1220. doi:10.1177/0269216319861921

3. Bruce A, Beuthin R. Medically Assisted Dying in Canada: "Beautiful Death" Is Transforming Nurses' Experiences of Suffering. *Canadian Journal of Nursing Research*. June 2019. doi:10.1177/0844562119856234

4. Alison A. Townsley. Putting a Bow on Death and Dying-Health Care Professionals' Experiences with Medical Assistance in Dying (MAiD): A Foucauldian Discourse Analysis with Agambian Insights. [Practice Based Research Paper on the Internet]. York University School of Social Work; 2018. [Cited February 28,2019] Available from: https://yorkspace.library.yorku.ca/xmlui/bitstream/handle/10315/35613/Alison_Townsley_PRP_2018.pdf?sequence=1

5. Gamondi C, Borasio GD, Oliver P, Preston N, Payne S. Responses to assisted suicide requests: an interview study with Swiss palliative care physicians. BMJ Support Palliat Care. 2019 Mar;9(1):e7. doi: 10.1136/bmjspcare-2016-001291. Epub 2017 Aug 11. Available from https://pubmed.ncbi.nlm.nih.gov/28801317/

OAIster included Results:

1. Implementing Medical Aid-in-Dying in Vermont: a Qualitative Study. *J GEN INTERN MED* **34,** 636–641 (2019). https://doi.org/10.1007/s11606-018-4811-1

2. Allyson Oliphant. "If Not Me, Then Who?" The Narratives of Medical Aid in Dying (MAiD) Providers and Supporters Around Their Professional Identity and Role in MAiD [Thesis on the Internet]. MacMaster University; 2017. [Cited on February 10, 2019]. Available from http://hdl.handle.net/11375/22146

BASE included results:

1. Sheridan, Laura, "When Patients Ask to Die: The Role of Nurses in Medical Assistance in Dying" (2017). Electronic Thesis and Dissertation Repository. 5041. https://ir.lib.uwo.ca/etd/5041

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Table 2: Description of articles included in qualitative meta-synthesis:

Study	Number and country of origin of participants	Description of participants	Extent of engagement in the MAiD process	Method of interview	Method of analysis	Emotional theme explored
1. Voorhees et al., 2014	23 physicians, 18 from USA (5 from Oregon), and 18 from Netherlands	@40% from primary care, majority >40 years	Physician assisted dying discussions.	40-70 min, one-one semi structured interviews	Modified 5-step framework-familiarization, identifying a theme, indexing, charting, mapping and interpretation.	Themes related to reflective emotions and sense of growth along with themes emotional labor and conscientious-based emotions.
2. Marwijk et al.,2007	22 primary care physicians, Netherlands	Variable range of experience, 5 PCPs participated in the Support and Consultation Regarding Euthanasia (SCRN)	Discussing and performing assisted death	4 focused groups, homogenized as per age and gender.	Content analysis within a coding frame of three themes of (1) emotional experience; (2) coping (dealing with and managing the event) and (3) role of the physician.	Themes related to reflective emotions and sense of growth along with themes emotional labor and conscientious-based emotions.
3. Denier et al., 2010	18 nurses from 5 provinces of Flanders, Belgium	Registered nurses (13 women, 5 men) of geriatric, oncology, internal medicine, and palliative care. All had positive attitude, except one who was conscientiously objecting.	Discussing and performing assisted death	1.5h in-depth interviews, think back to a specific, recent case of caring for a patient requesting euthanasia and to recount the way in which they experienced this	Grounded theory design	Themes related to role-assigned emotions along with themes of emotional labor.

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				process as a whole		
4. Norton et al., 2012	9 social worker hospice practitioners in Oregon, USA.	Represent several health systems in Oregon	involved in discussions with family of those participating in assisted death ('add on') and 'context interpreters'	Focused group	Thematic analysis	Themes related to role-assigned emotions (for example advocacy and feeling of being a 'gate-keeper')
5. Georges et al, 2008	30 general physicians in Netherlands.	71% male, 29% female, 46% had restrictive and 14% had permissive attitudes towards euthanasia.	89% had received explicit requests and were involved in discussions, and 64% had participated in EAS	In-depth interviews	Constant comparative method of analysis	Emotional theme of reflective emotions (example, feeling of sense of growth)
6.Snijde wind et al., 2014	28 General Physicians in Netherlands	Physicians who had received a request from someone suffering from dementia or a psychiatric illness, or who was ''tired of living,'' as these are cases that are often regarded as complex.	Involved in decision making of assisted death for respective patients.	In-depth interviews	Open coding and inductive analysis	Emotional theme of reflective emotions (example, reflecting on individual meaning of suffering)
7. Katja ten Cate et al., 2017	15 General Practitioners in Netherlands	8 GPs with liberal attitude, 5 with conservative attitude and 2 with neutral attitude towards assisted death. Mean age 51.2 years.	1-2/>2 assisted deaths performed.	In-depth interviews	several phases of coding (axial and selective coding); codes were refined, sub codes and overarching codes were assigned and relationships between codes were explored. Interviews were also analysed as a whole, to look for	Emotional theme of reflective emotions (example, reflecting on feelings of what is happening during the last stage of life)

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					patterns and inconsistencies in reasoning.	
8. Donald G Van Tol et al., 2012	15 physicians in Netherlands	Fourteen of them were general practitioners. Seven of them were also active as a consulting doctor, one was a nursing home doctor who was also working as a consulting doctor.	Physicians were consulting doctors of Euthanasia and have successfully completed a formal training program.	In-depth semi-structured interviews	Grounded theory approach by Glaser and Strauss and Glaser	Emotional theme of reflective emotions (example 'imagine self', cognitive reflection)
9. Melchor Lorraine 2018	8 social workers in California, USA.	75% female with 60% having an average 5 years of experience in hospice care.	assist patients and family with the death and dying process, may connect them to additional community resources, and offer counseling to improve and maintain emotional, psychological, social, and physical well-being	In-depth semi-structured interviews	Open coding, axial coding, selective coding, and conditional matrix stages of data analysis.	Emotional theme of role-assigned emotions (example, feeling of pro-self-determination and advocacy).
10. Miller et al., 2002	8 social workers in Oregon, USA	2 men, 6 women, age range of 27-64, 3-22 years' experience in hospice care	Active engagement in end-of-life care and assisted suicide discussions.	interviews	Ethnographic study and constant comparative method of analysis	Emotional theme of role-assigned emotions (example advocacy and self-determination)
11. Beuthin et al., 2018	17 Nurses in Canada	NPs, RNs, and LPNs, from urban and rural areas across Vancouver Island, British Columbia, working across	15 nurses had direct experience with MAiD, 7 were involved in some aspect of assisted death in the patient's journey (e.g., providing	In-depth semi structured interviews	Descriptive narrative enquiry and thematic analysis	Emotional theme of reflective emotions (example, a sense-making process)

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		settings including acute care, residential care, primary care clinics, and community and palliative care.	information, acting as witness to the medical assessment, providing care before or after, etc.)			
12. Bolt et al., 2016	8 pediatricians in Netherlands	8 pediatricians who were interviewed were 5 men and 3 women, aged 44–62y, working in four academic and three general hospitals .	25% had received an explicit request for Physician-assisted death, with 7% in the last two years, and the requests were mostly made by parents (25%) and sometimes by patients (6%)	Semi-structured interviews	Qualitative Analysis Guide of Leuven method was used for the analysis. Mixed method approach.	Emotional theme of role-assigned emotions (example, feeling of duty)
13. Dolares Angela Castelli Dransart et al., 2017.	1 physician, 8 directors of sociomedical institutions or organizations, 10 head nurses, 8 nurses, 10 nursing assistants or care assistants, and 3 sociocultural animators, Switzerland confronted with assisted suicide requests.	27 men, 13 women, mean age 52y.	14 had been faced with suicide or assisted suicide in their personal life, beside the situation of assisted suicide at work. None of the respondents interviewed had physically provided the lethal substance to perform the assisted suicide (a task assigned to Right to Die associations), nor were they directly involved in the decision-making process that enabled the assisted suicide to take place (except for one physician). In fact, the vast majority of these professionals	Semi-directive interviews conducted at workplace.	Grounded theory using 3 types of coding-open, axial and selective.	Emotional theme of role assigned emotions (example, feeling of professional compromise)

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			(except for two) declared that not only did they appreciate the fact that Right to Die associations assumed the task of delivering the lethal substance and physically assisting the requestor, but they also did not want to be led to do it themselves in the future			
14. Marianne Dees et al., 2012	28 physicians in Netherlands	20 males, 8 females, 22 GPs, 1 elderly care 2 GP trainees and 1 psychiatry	once in 3-5 years' experience with assisted death.	In-depth interviews with patients who had explicitly requested assisted death, their most involved relatives and their treating physicians	Thematic analysis	Emotional theme of reflective emotions (example, relational and feeling of trust in physician-patient relationship)
15. Harvath et al., 2006	20 hospice social workers and nurses in Oregon, USA.	--	The 20 hospice social workers/nurses described 33 different cases of terminally ill patients who had requested them to hasten death through physician assisted suicide (n = 22)	Semi-structured, In-depth interviews.	Thematic analysis	Emotional them of role-assigned emotions (example, feeling of professional failure, professional dilemmas and inner debate).

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16. Ina Otte et al., 2016	20 General practitioners (GPs) in Switzerland., 3 declined to participate due to personal discomfort with assisted death.	GPs who had chosen to refuse to assist a patient's suicide comprise the largest group in the study and provided the most insights.	Receive 1-3 requests of physician assisted suicide per year. 2/3rd of the GPs interviewed had chosen to refuse to assist a patient's suicide comprised the largest group in the study and provided the most insight into their handling of requests for PAS.	In-depth semi-structured interviews.	Thematic analysis	Emotional theme of basic emotions with conscience-based avoidance/rejection of MAiD (example, feeling of moral distress)
17. Ada van de Scheur and Arie van der Arend 1998	20 nurses in Netherlands	According to different phases of Euthanasia: Observation of a request for euthanasia: 17 nurses. 2) Decision making: 14 nurses. 3) Carrying out of euthanasia: 12 nurses. 4) Aftercare: 14 nurses	Engagement as per different phases of Euthanasia	In-depth semi-structured interviews.	Thematic analysis	Emotional theme of role-assigned emotions (example, feeling of moral distress)
18. Emmanuelle Bélanger et al., 2018	18 university affiliated palliative care physicians in Quebec, Canada	Participants positioned themselves opposite euthanasia	majority of the palliative care physicians on staff at the palliative care units of two public hospitals located in an urban area of Quebec. All participants were full-time palliative care physicians, and like most palliative care providers in	In-depth semi-structured interviews.	Inductive methodology of Interpretive description.	Emotional theme of role-assigned emotions (example, professional dilemmas and conflicting values with palliative care)

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			Canada, the majority of them (16 out of 18) were family physicians. As expected, all participants expressed discomfort with euthanasia as an aspect of end-of-life care. All but one denied the influence of religious or political positions in shaping their views.			
19. Jessica Shaw et al., 2018	Eight physicians who offered MAID in British Columbia in 2016, Canada	3 were from greater Vancouver, 3 were from Victoria, and 2 worked in a small community on Vancouver Island. Seven were family doctors and 1 was a general internist. Their ages ranged from 37 to 64 years. There were 2 men and 6 women; 6 worked full-time and 2 worked part-time.	Collectively, by the end of December 2016, the 8 physicians in this study had assessed 332 people who were seeking MAID and had completed 135 assisted deaths	In-depth semi structured interview via phone call/email	Qualitative thematic analysis	Emotional theme of basic emotions, especially positive emotions (example, sense of fulfilment)
20. Judith Schwarz, 2004	10 nurses who worked in home hospice, critical care, and HIV/AIDS care settings, USA	Four worked in hospice home care, three were advance practice nurses who worked with persons with AIDS, two worked	Nurses were eligible to participate in this study if they believed that a competent patient had made a serious request for their help in dying.	In-depth interviews done at least twice for 7 participants .	van Manen's approach to phenomenology phenomenological interpretation and analysis (phenomenological enquiry)	Emotional theme of role-assigned emotions (example, feeling of human-human response and connectedness)

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		in critical care, and one was a clinical nurse specialist in the care of patients with spinal cord injuries. Two of the ten nurses were male, all were Caucasian, middle-aged, well educated (three PhDs; five Masters of Science in Nursing), and clinically experienced (6–35 years)				
21. Marie-Eve Bouthillier and Lucie Opatrny 2019	22 conscientiously objecting physicians in Quebec, Canada	26 to 67 years (mean: 45 years), 12 of them were male (54.5%). 14 Family physicians, 2 oncology and 1 each from psychiatry, neurology, nephrology, intensive care, geriatrics and pneumology. 14 from catholic background.	Physicians had received requests, had discussions with patients regards to MAiD, and conscientiously objected to participate.	Semi-structured interviews. eight open-ended questions Interviews ranged in length from 15 min to 1 h, with a mean length of 24 min (median length = 21 min). think back to their first medical aid in dying request (as some physicians had received	descriptive thematic analysis	Emotional theme of basic emotions (for example emotional labor, burden and fear of psychological repercussions)

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				more than one request) and describe the reasons which motivated their refusal.		
22. Gamondi et al., 2017	23 palliative care physicians across Switzerland	65% German, 30% French and 5% Italian speaking	Regularly received assisted suicide requests. The involvement of Swiss physicians is mostly confined to the decision-making phase; medical certification of diagnosis and mental capacity.	Semi-structured interviews.	thematic analysis	Emotional theme of role-assigned emotions (example professional role-related feeling of ambiguity, fear of being stigmatized as physicians, feeling of walking a tight rope.)
23. Rosanne Beuthin, 2018	female, of Anglo-European ancestry, age mid-fifties, living in an urban center, Canada	Doctorate in nursing and was employed as a consultant under an end-of-life Program to enact a new MAiD program.	daily journal entries made over a 6 month period, from the first day of immersion in the role and culture of MAiD from late May to October 2016	Raw autobiographical text held scattered floods of ideas and released emotions into a thick created Story.	autoethnographic approach-reflective analysis	Emotional theme of reflective emotions (example, feeling of embodiment, compassionate care and sense-making reflective emotions. Exploring tensions around language, attitudes)
24. Anne Bruce and Rosanne Beuthin, 2019	15 RNs/NPs/LPNs from British Columbia, Canada.	Participants worked in diverse settings including acute care, community-home care, and specialty areas including emergency	Eight nurses had directly aided with MAiD and cared for the patient at home or in a care setting. Seven had been involved indirectly with patients such as providing assisted	Semi-structured interviews-(1) tell me about your first experience of being asked to participate	narrative inquiry and thematic analysis	Emotional theme of reflective emotions (example fear of desensitization with deeper questioning) along with complex emotions of "compassion

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		room and palliative care.	dying information upon request and listening to patients and families as they explored pursuing MAiD	in a medically assisted death and how you came to the decision to participate or not and (2) tell me about the MAiD experience itself. What was most challenging ?		satisfaction" as well as compassion fatigue
25. Alison Townsley 2018	seven nurses, social workers, and personal support workers, Canada	Health care professional enrolled through purposive sampling.	Engaged in discussions and assessments of patients requesting MAiD.	one-on-one, semi-structured interviews with health care professionals	Foucauldian Discourse Analysis perspective. Interview data is analyzed by situating the health care professional as an effect, as a producer, and as a challenger of power-knowledge systems. Philosophical theories of Giorgio Agamben are applied to the data to challenge Foucauldian principles, and to bolster the discussion of defining of the body that deserves to live,	Emotional theme of reflective emotions (example, emotions emerging from engagement of the individual in terms of power, knowledge and individual identity)

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					and the body that deserves to die.	
26. Buchbinder et al., 2019	37 health care providers in Vermont, USA.	Health care providers from Hospital and community-based practices. Most were women (68%) and the largest subgroup specialized in internal or family medicine (53%). Most of the nurses and social workers were women (89%) and most worked for hospice and home health agencies (61%).	19 physicians (10 internal medicine, 4 palliative care, 3 neurology, 2 oncology), 12 had participated in Act 39 (The patient Choice and control at End-of-Life Act) as prescribing physicians, the remainder had initiated but not completed the Act 39 protocol (n = 3), participated as a second physician to confirm the patient's diagnosis, prognosis, and decisional capacity (n = 3), or counseled patients (n = 1). The mean age of nurses and social workers (n=18, 9 hospice/home nurse, nurse practitioner 5, inpatient palliative care 2, hospice social worker 2) was 52.5, with most working for hospice and home health agencies (61%). While all professionals in this group engaged in clinical care for patients pursuing Act 39, specialty clinic nurse	One-to-one semi structured interviews	Grounded theory approach	Emotional theme of role-assigned emotions (example pride, burden etc.)

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			practitioners were more likely to assist with navigating access to the aid in dying. Participating health care professionals worked in ten of Vermont's 14 counties			
27. Allyson Oliphant, 2017	4 physicians. 4 nurses and 6 HCPs (allied health care professional social workers (1), spiritual care providers (1), pharmacists (1), genetic technologists (1) and psychologists (2).) of team ADRAS in Hamilton, ON.	Of the data available, 2 were semi-retired family physicians, One is an intensive care physician with a background in cardiology, and the second is an Emergency Room physician with training in palliative care.	All participants are members of the ADRAS (assisted dying resource and assessment service) who support the practice of MAiD. Every participant had a capacity to be flexible.	One to one semi-structured interviews.	Grounded theory approach	Emotional theme of reflective emotions (example, emotions related to related to professional identity, sense making, feeling of obligation to serve)
28. Laura Sheridon 2017	nine palliative care nurses in southwestern Ontario, Canada	3 males, 6 females. 3 participants worked in residential hospices where MAiD was not supported as an end-of-life option, six participants worked in the community providing home care where MAiD is an option in end-of-life planning. Two participants had	Participants in the study indicated that nurses may act as a liaison between physicians and nurse practitioners who have the authority to assess patient eligibility and provide the intervention of MAiD and the patient, notifying them of an inquiry about or a request for MAiD	One-to-one semi structured interview.	interpretive description qualitative methodology	Emotional theme related to role-assigned emotions (example, emotional expressions ("hard conversations") related to nursing role, struggle related to moral conflicts.

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		previous inpatient hospital experience in emergency care and in intensive care specialties.				
29. Khosnood et al., 2018	19 physicians, Canada. Quebec not included.	Half of the participants were palliative care specialists (n = 8), with the remaining representing Family Medicine (n = 4), Anesthesia (n = 2), Hematology (n = 1), and Obstetrics & Gynecology (n = 1). The majority of participants practiced in an urban setting (n = 13).	Average 6.9 MAiD cases.	In-depth semi-structured telephone-based interviews.	inductive thematic analysis approach	Emotional theme of role-assigned emotions (example burn out, negative effect on inter-professional relationships vs. increased feeling of respect)
30. Beuthin et al., 2020	8 physicians, Canada.	Participants included general practitioners (GPs) and Non-specialist physicians from urban and rural communities working in acute and palliative care. Ages ranged from 33 to 62 years (average age 49), with an equal number of men and women. The majority identified no active religious	experience with MAiD provision ranged from 12 to 113 assisted deaths. Only one physician was dedicated to full-time provision.	In-person or telephone-based semi-structured interviews.	interpretive descriptive methodology and thematic analysis	Emotional them of reflective emotions, (example complex emotions of compassion satisfaction, embodied awareness, soul-searching)

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		affiliation, and ethnicity was withheld to protect anonymity. Years of experience ranged from 6 to 38 years (average of 23).				
31. Keri-Lyn Durant and Katherine Kortes-Miller 2020	23 physicians of Rural area, northwestern Ontario, most of subarctic Ontario.	23 physician participants ranged in age from 26 to 63, with a mean age of 43 years. Physicians worked in a variety of settings, with 14 in an urban setting – in family practice, as a hospitalist or other specialist, in the emergency department, in palliative care, and in long-term care. Nine participants declared a rural practice, and self-identified as rural generalists, working on a First Nations' reserve, in a community, at a satellite clinic, or 'All of the above'.	11 identifying themselves as acting both as assessor and provider, 1 as assessor only, 4 as providing referrals upon request, and 7 without any direct/indirect experience. These seven were included in the study because they expressed a desire to participate and reported that their practice and the community had been impacted by the legislation. There was also a variance in terms of exposure to death in practice, with an estimated total between 2 and 250 deaths per annum	using 1 semi-structured focus group and 18 semi-structured interviews comprising 9 set of questions	Thematic analysis	Emotional theme of role-assigned emotions (example, feeling of impact on inter-professional relationships, feeling of unpreparedness.
32. Snijdewind et al., 2016	secondary analysis of in-depth	Respondents were recruited both by the network of physicians	Twenty-two respondents worked as family physicians, and six	One-to-one semi-structured interviews.	Thematic analysis	Emotional theme of reflective emotions (example, those related to meaning

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	interviews with 28 Dutch physicians who had experience with a complex case of EAS	working for SCEN (Support and Consultation for Euthanasia in the Netherlands) as well as via a national Questionnaire. Nine of the respondents were female. The respondents' age ranged from 36 to 68 years	worked as medical specialists (three elderly care physicians, a psychiatrist, an internist and a lung specialist). Next to this, six of the respondents also worked as SCEN physicians. All had experience with EAS requests and the performance of EAS.			of suffering, blurring emotional boundaries)
33. Pesut et al., 2020	59 registered nurses and nurse practitioners in Canada	n = 9 (15%) were conscientious objectors, Spiritual or Religious Affiliation: n = 33 (56%) Neither: n = 15 (25%); Spiritual but not Religious: n = 11 (19%) Home & Community: n = 32 (54%); Acute Care: n = 10 (17%); Long-term care: n = 5 (9%); Hospice: n = 4 (7%); Clinic: n = 3 (5%	24 of the 59 participants had conducted more than 25 conversations with patients about MAiD, and 11 of the 59 participants had been involved with more than 25 patients who went on to receive MAiD.	Semi-structured interviews conducted on telephone. Question examples: (i) Can you tell us how the process of MAiD occurs in your practice context? (ii) What resources and practice supports are available to assist you in caring for MAiD patients? (iii) Tell us about your experiences with MAiD?	Qualitative approach guided by Interpretive Description. data immersion, open coding, constant comparative analysis, and the construction of a thematic and interpretive account. Transcripts include emotions evident during the interview (e.g., crying).	Emotional theme of role-assigned emotions (example, emotions related to find themselves caught between the proverbial "rock and hard place." With feelings of Emotions of frustration, powerfulness of the experience, feeling drained out)

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				The average length of interviews was 55 min.		
34. Deborah Volkar et al., 2001	40 oncology nurses who received requests for assisted death in USA.	48% in hospital/multi-hospital settings. 9 female, 1 male. Mean age 45 y.	30% had received requests for assisted suicide, 6 (1%) engaged in assisted suicide, and 20 (4.5%) admitted to intentionally injecting a drug to end a patient's life.	Recipients were re quested to submit a written account or story of receiving a request for assistance in dying from a terminally ill patient with cancer.	Denzin's process of interpretive interactionism with an emic, ideographic approach. That is, individual experience is considered to be unique; discovery of an individual's epiphany and associated meanings is the research focus	Emotional theme of basic emotions (example emotional labor) along with reflective emotions of feeling lack of control (or lack of it) and moral distress).
35. Mathews et al., 2021	23 palliative care providers (13 physicians and 10 nurses) who practiced for 6 months or more before and after the introduction of MAiD, in inpatient and community-based settings that supported assisted death in southern Ontario, Canada.	54% of physicians and 90% of nurses were female with a mean age of 43 years and 42.6 years respectively.	All the participants described having discussions with patients regarding MAiD and 7/23 participants (4 nurses and 3 physicians) described directly witnessing assisted death. 8/13 physicians made referrals for MAiD, 4 conducted assessments, and 3 physicians were MAiD providers; 3 physicians identified as conscientious objectors. None of the nurses identified themselves as conscientious	Semi-structured interview based on pre-determined interview guide	Braun and Clarke's version of Thematic analysis	Emotional theme of role-assigned emotions (example Role-driven emotional themes of Emotional, psychological and resource burden along with theme of emotional labor)

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			objectors, although some expressed moral or religious conflict around MAiD.			

Table 3: Codes and Themes table: This table represents line-by-line coding (underlined) of each study (numbered in parenthesis corresponding to the table 2 above). These codes have been subsequently grouped into descriptive themes in their respective boxes.

A) Over-arching theme of basic emotions:

Theme 1: Emotional labor (positive/negative emotions)	Theme 2: Conscience based emotions.
Codes: "rewarding" "liberating", "Well please let someone else do this question", "blood had frozen in my veins", I just felt just totally cold all over. I had no idea of what to do. I realized there was no help I could get from anywhere . . . I . . . felt as though I was . . .impotent to help them. "If possible, I would run away. But I see it as the last part of my care. I have taken care of that patient for years and now at the moment . . . when she needs me most . . . I would be a coward to run away then. (1) "I felt very lonely" "heroic feelings", "tense", "scary", "terribly creepy", "felt pressured to succeed", "suffer a loss yourself when someone like that dies" "terribly manipulated", "felt slightly put upon, angry" 'let off steam' (2) "feeling of ambivalence", "intense", "gradually feel less secure, less fearful", "surprisingly grateful" . "very demanding and emotionally distressing" (3) "very demanding, generally like to avoid", "drastic"(5) , "moral pressure" , "uncertain, complex"(6), "very hard"(7), "feeling choked up or shedding a tear" "Feeling positive emotions of peace and amazement were more surprising and often shared cautiously in public" ,"had difficulty finding effective words for the paradoxical experience of witnessing death that is, both "sad" and "beautiful." (11). "felt reluctant as it is difficult to predict" (12). "feeling of enrichment", "feeling of sorrow and intrusive thoughts", "feeling like weathering the storm", "empathy and emotional closeness", "personal compromise" (13). "do not feel competent" (16).	Codes: "making pluses and minuses about it . . . but . . . 'What's it doing to me? I'm going to kill someone tonight.'[respondent began to cry], "I have to do no harm, and I just feel that if you're assisting someone in dying . . . it's against what I've been trained . . . It's not up to me to decide when the patient dies . . ." (1); "killing another person is not the solution. It's in the ten commandments" "sense of guilt. I feel as if I'm an executioner. Who am I to have the right to do this?" (2); "Conscientiously, I find it hard to come to terms with euthanasia" (3); Clarity of conscience- "a sort of trap that can't be avoided. That in spite of everything you can offer, a terminal stage can be so heavy, perhaps too heavy for a patient. In fact, I always see it as an emergency exit. When I am talking about it with a patient I say, "yes we will consider it, if you don't want to go on any longer and if I have nothing more to offer you to make it better"(5); "I am a Christian so I have strong feelings because of my belief and my background, believe that no human being should be in the position to hasten death." (10); cannot bear the idea of killing one of my patients", I do not feel competent to deal with the topic…especially for my personal psychological health, "challenges my belief, I do not understand how it can be meaningful" (16)

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"rewarding work", "honor", "bit overwhelming", "proud", "incredible" "feeling like being on call all the time (19), "emotional burden", "fear of psychological repercussions", "uncomfortable", fear of stigmatization (21), "fear of stigma/isolation, feeling of ambiguity" (22), "feeling courageous" (23), "satisfying and gratifying" " roller coaster" , "transformational feelings of beautiful death" (24), just feel coldness, or whatever. You just feel drained …"(28), "unexpected rewards" , "enriching capacity of caring", (30), , "anxiety, shock, self-doubt", "deep inside…conflict" (34); "walking quiet a tight rope", was as prepared…but went outside and felt like I was about to throw up", "actually, find them. . . they're such beautiful experiences with family. It's the shared experience with the family that you're with that you have an opportunity to help." (35)	"to see somebody lying there, to whom you brought a cup of tea that morning. And you know that everybody who gets a heart attack can die as well, but this was no heart attack. You know that, of course. So, somebody has been killed, just like that. . . That makes it different." (17) "conflicted, trying to reconcile their own personal moral stance with facilitating the end of someone's life" (28) "What would my family think that I'm working on a unit that does that [Medical Assistance in Dying]? Do I hide it from them. . .what if people find out that we do it? Are people going to come up here and start protesting? People will see that as evil." (35)

B) Overarching theme of reflective emotions.

Theme 1: relational "feeling of trust and sympathy in physician patient relationship strong" (14) "human centered, compassionate care" (23), "for somebody to approach you is almost an honor that they trust you enough to have this conversation, and to have to sort of shut them down, or acknowledge how they're feeling" (empathy) (28), "intimate, emotional engagement-rediscovering the art of medicine", (30), "indelible nature of the experience shared" (34) "as soon the topic [Medical Assistance in Dying] came up, that I was a conscientious objector and the person said that you're not on my side, even though she was getting the service [MAID] . . .I was seen as somebody who was not helping her" (35)	Theme 2: Discourse based (control over a natural process of dying) "interesting discourse presented itself through idea of using stages to determine someone's chances of survival, and the need for professionals to have something finite and concrete to measure", "discourse that emerged through conversations with participants was how control (or masterhood) equates to people's sense of wellbeing" "MAiD itself presents a paradox insofar as one can be too sick to access this form of assistance that is exclusively designed to bring death to the most critically ill people" "The most dominant discourse that emerged from this data set was participants aligning what is right and good within the confines of the law." (25); "medicalization of a social problem" (32); "degree of control over dying process" (34).
Theme 3: Sense making process and related emotions. (Theme of Growth) "You grow with the problems of the patients" (1) " stay closer to their own beliefs" " long road to becoming aware of one's own views" (2) "meaning full experience" " almost closer than when someone is having a baby" (5) ''[EAS] is not an act, it's a process towards which we both grow" (6) , "Being in process, holding an in-between space of uncertainty, reflection, and active sense-making" (11); "pure moment of autonomous self-consciousness" "I am working and sense making as I go along, being sure that I keep breathing",	Theme 4: Process influenced themes (suffering---relief--death) "Invisible suffering made it harder for the people close by to empathize and come to terms with the patient's request and his/her death" (6); "for me, a lot of talk, talk about death and dying, talk about life, about saying goodbye, really seeing and feeling what is happening in this last phase of life and reflect on that. But not everybody is capable of talking and reflecting this way, while everybody is going to die. So that's my problem" (7); "imagine self" and "imagine other" cognitive route. Use of cognitive reflection (8);

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"feeling of embodiment, become the face of MAiD", "bearing witness"(23); "worries of becoming desensitized and ongoing deeper questioning" (24); "their thoughtful silence after speaking or listening represented and solicited from me respect for the dead and the dying, seething inner anger, and perhaps the quietude that one experiences when their physical body feels the effects of being a challenger and resister in the strongest way possible" "Kind of letting them have control over what they can have control over" "beautiful journey of self-reflection", "grappling with identity" (25); "embodied awareness", "soul searching" (30); "silent knowing" (34)	"very difficult for me to let…go, to be so aware of saying farewell, and now I notice that as time passes it gets harder and harder for me" (14); "sense of urgency to hasten death" (23); "boundaries of EAS has shifted over time, making feel stretched, tense and insecure" "not feeling competent if suffering is existential" (32); "it's been a bit of a challenge to delineate what we're doing in relationship to the request for assisted dying and what normal care still continues to be" "struggle with the rules of a complex legislated and reporting process that determines it"(33)

C) Overarching theme of emotions related to professional values:

Theme 4: Role-assigned emotions **Nurses:** "predominantly tend to be conformist (following existing conventions rather than using critical reflection) when faced with ethical dilemmas. Combined with the emphasis of the medical responsibility in euthanasia care, and combined with the strong inclination of nurses to respect the patients' wishes, it seems logical that nurses interpret the gravity of the process in emotional terms"(3); ""unchartered territory," where "there was almost no foundation" for providing this option, and "this is a whole new role for all of us.(being pioneers)" "duty to provide care" is being touted as "you don't have a choice" and the information isn't there [about] how to object if you don't agree with" (11); "moral distress", "burden", see somebody lying there, to whom you brought a cup of tea that morning. And you know that everybody who gets a heart attack can die as well, but this was no heart attack. You know that, of course. So, somebody has been killed, just like that. . . That makes it different" (16); "identifying the moral line", "human-human response and connectedness because of the role played", "fear the potential for abuse, and the possibility that other health-care professionals might too readily accept a patient's fleeting wish to die" (20); "taken for granted, feeling terrible" "their own suffering is invisible" (24); "walking alongside patients" like the experience of being able to make [death] a better experience. That celebration of life rather than the mourning of death" (27); "feeling of having hard conversations" (28); "Nurses seeking to provide the compassionate care consistent with such a momentous moment in patients' lives, without suitable supports, find themselves caught between the proverbial rock and hard place" "powerful experience" "mad as a hell", "overwhelmed" "…don't find the provisions so emotionally draining, but it's more the logistics and it's a lot of work as a nurse" (33); there's a sense of ceremony [before Medical Assistance in Dying], So, those all have impacts in terms of resources" (35). **social worker**: "feeling of being a gatekeeper" (4); "sense of preparedness", feeling that this option is 'pro-self-determination which is our job"(9); "inner debate, cannot make peace with that, felt a huge shift in my ethics", "dying process has a lot to give" "missed opportunity to deepen oneself spiritually", "missed opportunity to forgive"(15); feeling of advocacy and self-determination in sync with hospice and social work values, and we will advocate for the patients . . . to get them whatever they want . . . I believe in self-determination, but I think it's (PAS) a sad commentary on our society." "Our job is to meet the patients where they are" (10); "felt like higher commitment", "felt like a failure if patient chose EAS" (16). **physicians**: "heavy responsibility" (5); "implicit ethical tension due to pressure to decide", "It is the right time for EAS] Only if someone is totally at peace with himself, his life and his death, and if I see and feel that too.'(7); "feeling of duty" (12);

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"professional compromise" (13); "fears prosecution", "burden, not wanting to abandon the patient" (14); extremely personal choice, I feel that part of some of the resistance at times can be related to [the fact that] that this is something new" "feeling of being torn between professional values and patient values (18); "significant administrative burden" (21); "struggle to reconcile to professional values", sense of responsibility to not create barriers" "walking a tight rope" (22); "tremendous pride", "burden as well" (26); duty to serve. "if not me than who" (27); "interprofessional lack of trust" "excessive workload and lack of financial satisfaction" (29); "burgeoning relationship between palliative care and MAiD", " positive because master of destiny" , "uncomfortable discussing it" (31); "Good palliative care takes a lot of time and interdisciplinary resources. . .when a patient is requesting MAID, most of the resources have been sucked up by that one case … And all of the high-quality palliative care that we do falls by the wayside for the other patients." (35)

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Supplementary appendix 4: PRISMA and ENTREQ checklist.

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	p.4-6
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	p.2 and p.6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	p.6-8
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	p.8-9, supplementary appendix 1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Supplementary appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	p.9
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	p.9-10
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	p.6-7
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	p.6-7
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	p.9-10, supplementary appendix 2
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Not applicable
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	p.10-11
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	p.10-11
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	p.10-11 supplementary appendix 3
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	p.10-11
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Not applicable
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Not applicable

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Supplementary appendix 4: PRISMA and ENTREQ checklist.

Section and Topic	Item #	Checklist item	Location where item is reported
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	high risk, p.20
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	p.5, supplementary appendix 3
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1, p.7
Study characteristics	17	Cite each included study and present its characteristics.	Supplementary appendix 3
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supplementary appendix 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Not applicable
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Supplementary appendix 2
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	p.11-14. Supplementary appendix 3
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Not applicable
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Not applicable
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Not applicable
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 1
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	p.17-20
	23b	Discuss any limitations of the evidence included in the review.	p.17-20
	23c	Discuss any limitations of the review processes used.	p.17-20
	23d	Discuss implications of the results for practice, policy, and future research.	p.17-20
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	p. 11
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	p.11
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	none
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	p.21
Competing interests	26	Declare any competing interests of review authors.	p.21

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Supplementary appendix 4: PRISMA and ENTREQ checklist.

Section and Topic	Item #	Checklist item	Location where item is reported
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	p.21

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71
For more information, visit: http://www.prisma-statement.org/

Enhancing transparency in reporting the synthesis of qualitative research: ENTREQ Checklist (Tong, *et al.,* 2012)

Item No.	Guide and Description	Report Location
1. Aim	State the research question the synthesis addresses	Background, p.6
2. Synthesis methodology	Identify the synthesis methodology or theoretical framework which underpins the synthesis, and describe the rationale for choice of methodology (e.g. meta-ethnography, thematic synthesis, critical interpretive synthesis, grounded theory synthesis, realist synthesis, meta-aggregation, meta-study, framework synthesis)	Data analysis, p.10
3. Approach to searching	Indicate whether the search was pre-planned (comprehensive search strategies to seek all available studies) or iterative (to seek all available concepts until they theoretical saturation is achieved)	search strategy screening and eligibility criteria *SPIDER,* p.6
4. Inclusion criteria	Specify the inclusion/exclusion criteria (e.g. in terms of population, language, year limits, type of publication, study type)	*Eligibility criteria,* p.7
5. Data sources	Describe the information sources used (e.g. electronic databases (MEDLINE, EMBASE, CINAHL, psycINFO), grey literature databases (digital thesis, policy reports), relevant organisational websites, experts, information specialists, generic web searches (Google Scholar) hand searching, reference lists) and when the searches conducted; provide the rationale for using the data sources	search strategy, p.8
6. Electronic Search strategy	Describe the literature search (e.g. provide electronic search strategies with population terms, clinical or health topic terms, experiential or social phenomena related terms, filters for qualitative research, and search limits)	Supplementary appendix 1 and p.6-9
7. Study screening methods	Describe the process of study screening and sifting (e.g. title, abstract and full text review, number of independent reviewers who screened studies)	p.9 study selection process, *Fig 1 PRISMA flow diagram*

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Supplementary appendix 4: PRISMA and ENTREQ checklist.

8. Study characteristics	Present the characteristics of the included studies (e.g. year of publication, country, population, number of participants, data collection, methodology, analysis, research questions)	*Table 2 in supplementary appendix 3, Characteristics of included studies*
9. Study selection results	Identify the number of studies screened and provide reasons for study exclusion (e.g. for comprehensive searching, provide numbers of studies screened and reasons for exclusion indicated in a figure/flowchart; for iterative searching describe reasons for study exclusion and inclusion based on modifications to the research question and/or contribution to theory development)	Fig 1 - PRISMA flow diagram
10. Rationale for appraisal	Describe the rationale and approach used to appraise the included studies or selected findings (e.g. assessment of conduct (validity and robustness), assessment of reporting (transparency), assessment of content and utility of the findings)	Table 1, CERQual approach
11. Appraisal items	State the tools, frameworks and criteria used to appraise the studies or selected findings (e.g. Existing tools: CASP, QARI, COREQ, Mays and Pope [25]; reviewer developed tools; describe the domains assessed: research team, study design, data analysis and interpretations, reporting)	Appraisal of the methodological limitations of included studies, Table 1, CERQual approach
12. Appraisal process	Indicate whether the appraisal was conducted independently by more than one reviewer and if consensus was required	p.10, independently done by the three researchers and consensus achieved.
13. Appraisal results	Present results of the quality assessment and indicate which articles, if any, were weighted/excluded based on the assessment and give the rationale	Table 1, CERQual approach
14. Data extraction	Indicate which sections of the primary studies were analysed and how were the data extracted from the primary studies? (e.g. all text under the headings "results /conclusions" were extracted electronically and entered into a computer software)	Data extraction and analysis, p.10
15. Software	State the computer software used, if any	None used
16. Number of reviewers	Identify who was involved in coding and analysis	
17. Coding	Describe the process for coding of data (e.g. line by line coding to search for concepts)	p.10
18. Study comparison	Describe how were comparisons made within and across studies (e.g. subsequent studies were coded into pre-existing concepts, and new concepts were created when deemed necessary)	Table 2 in supplementary appendix 3.
19. Derivation of themes	Explain whether the process of deriving the themes or constructs was inductive or deductive	Inductive process, p.10

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Supplementary appendix 4: PRISMA and ENTREQ checklist.

20. Quotations	Provide quotations from the primary studies to illustrate themes/constructs, and identify whether the quotations were participant quotations of the author's interpretation	p.12-13
21. Synthesis output	Present rich, compelling and useful results that go beyond a summary of the primary studies (e.g. new interpretation, models of evidence, conceptual models, analytical framework, development of a new theory or construct)	Discussion, p.17-20

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